MOG JEE Main Full Test - 2024 Revised

Total Time: 180 minutes | Total Marks: 300 | Total Questions: 75

GENERAL INSTRUCTIONS

The Question Paper consists of three parts (Physics, Chemistry and Mathematics) with a total of **75** questions.
Each subject is divided into 2 sections:
Section 1: Single Correct Type Questions (4 options, only one correct)
Section 2: Numerical Value Type Questions (answer to be entered as numerical value)
Marking Scheme:
Single Correct: +4 for correct, -1 for wrong, 0 if unattempted.
Numerical Value: +4 for correct, -1 for wrong.

By: C I P H Ξ R

Q.1

Proton striking a stationary lithium target activate a reaction : $\mathrm{Li}^7 + \mathrm{p} \rightarrow \mathrm{Be}^7 + \mathrm{n}$
At what value of proton's kinetic energy can the resulting neutron be stationary.
Given: $\mathrm{m}\left({}_1^1\mathrm{H}\right) = 1.00783 \text{ amu}, \mathrm{m}\left(\mathrm{Be}^7\right) = 7.01693,\ \mathrm{m}\left(\mathrm{Li}^7\right) = 7.01601,\ \mathrm{m(n)} = 1.00867, 1 \text{ amu} = 931 \text{ MeV}$

(A) 0.2 MeV

(B) 1.9 MeV

(C) 3.8 MeV

(D) 7 MeV

Q.2

A body is thrown horizontally from the top of a tower and strikes the ground after three seconds at an angle of 45° with the horizontal. Find the speed with which the body was projected.
(Take, $g = 9.8 \text{ m s}^{-2}$)

(A) 31.8 m s^{-1}

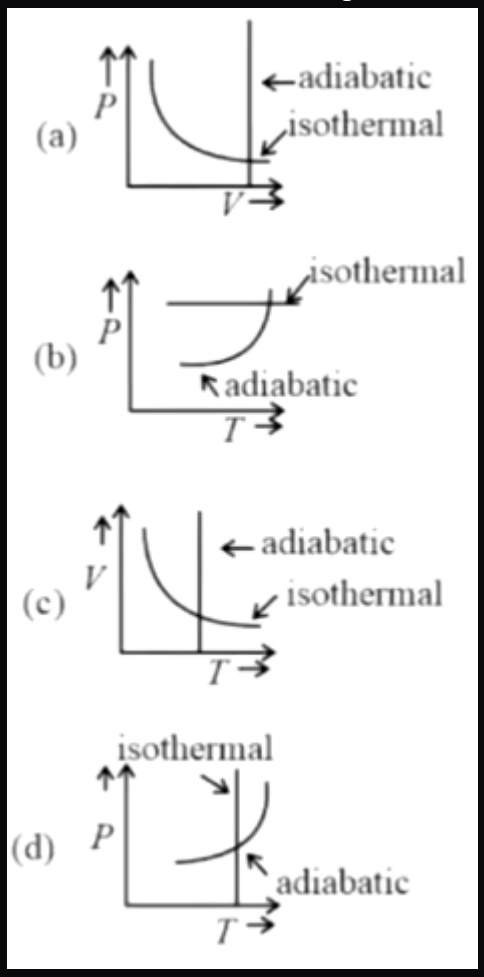
(B) 25.6 m s^{-1}

(C) 29.4 m s^{-1}

(D) 26.7 m s^{-1}

Q.3

Which one is the correct option tor the two different thermodynamic processes ?



(A) (c) and (a)

(B) (c) and (d)

(C) (d) only

(D) (b) and (c)

Q.4

The electric field part of an electromagnetic wave in vacuum is
$E = 3.1\ \mathrm{N\ C^{-1}} \cos\left[(1.8\ \mathrm{rad\ m^{-1}})y + (5.4 \times 10^8\ \mathrm{rad\ s^{-1}})t\right]\hat{\mathrm{i}}$
The wavelength of this part of electromagnetic wave is,

(A) $1.5\ \mathrm{m}$

(B) $2\ \mathrm{m}$

(C) $2.5\ \mathrm{m}$

(D) $3.5\ \mathrm{m}$

Q.5

A simple pendulum has a time period T_1 when on the Earth's surface and T_2 when taken to a height R, above the Earth's surface, where R is the radius of the Earth. The value of $\frac{T_2}{T_1}$ is

(A) 1

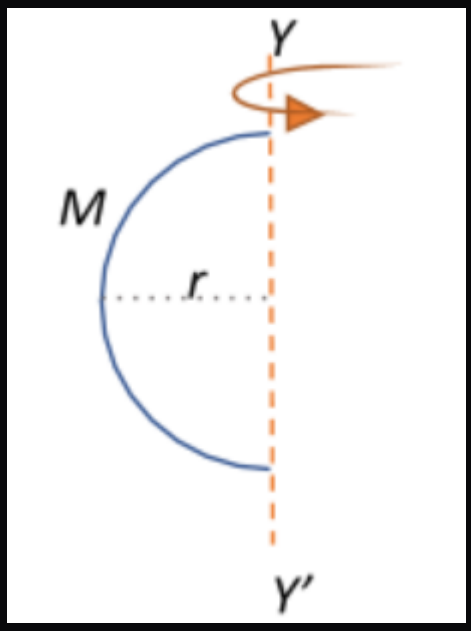
(B) $\sqrt{2}$

(C) 4

(D) 2

Q.6

A thin wire of length l and mass m is bent in the form of semicircle. Its moment of inertia about an axis joining its free ends will be



(A) ml^2

(B) zero

(C) $ml^2/2\pi^2$

(D) None of these

Q.7

A wire of length 2 m is clamped horizontally between two fixed support. A mass $\mathrm{m} = 5$ kg is hanged from middle of wire. What would be its vertical depression (in cm) in equilibrium?
(Young modulus of wire $= 2.4 \times 10^9$ N m^2, cross-sectional area $= 1$ cm^2)

(A) 5.92

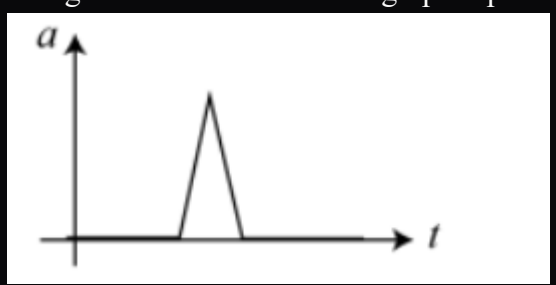
(B) 4.84

(C) 4.64

(D) 3.84

Q.8

The given acceleration -time graph represents which of the following physical situations?



(A) A cricket ball moving with a uniform speed hit with a bat for a very short time interval

(B) A ball is falling freely from the top of a tower.

(C) A car moving with constant velocity on a straight road.

(D) A football is kicked into the air vertically upwards.

Q.9

Assertion: In S. H. M., the velocity and displacement of the particle are in the same phase.
Reason: Velocity is the ratio of displacement to the time taken.

(A) Assertion is True, Reason is True; Reason is a correct explanation for Assertion

(B) Assertion is True, Reason is True; Reason is not a correct explanation for Assertion

(C) Assertion is True, Reason is False

(D) Assertion is False but Reason is True.

Q.10

Diameter of a steel ball is measured using a Vernier cllipers which has divisions of $0.1\ \text{cm}$ on its main scale (MS) and 10 divisions of its Vernier scale (VS) match 9 divisions on the main scale. Three such measurement for a ball are given as.

S. No.	MS (cm)	VS divisions
1.	0.5	8
2.	0.5	4
3.	0.5	6

If the zero error is $-0.03\ \text{cm}$, then mean corrected diameter is.

(A) $0.53\ \text{cm}$

(B) $0.56\ \text{cm}$

(C) $0.59\ \text{cm}$

(D) $0.52\ \text{cm}$

Q.11

In an experiment, the percentage of error occurred in the measurement of physical quantities $A,\ B,\ C$ and D are 1%, 2%, 3% and 4%, respectively. Then the maximum percentage of error in the measurement X, where $X = \frac{A^2 B^{\frac{1}{2}}}{C^{\frac{1}{3}} D^3}$ will be

(A) $\left(\frac{3}{13}\right)\%$

(B) 16%

(C) -10%

(D) 10%

Q.12

Assertion: Interference can be obtained by using two different lamps.

Reason: Two different lamps are incoherent sources as constant phase difference cannot be maintained between them.

(A) If both Assertion and Reason are true and the Reason is correct explanation of the Assertion.

(B) If both Assertion and Reason are true but Reason is not explanation of the Assertion.

(C) If Assertion is true but the Reason is false.

(D) If Assertion is false but Reason is true.

Q.13

The bodies of masses 100 kg and 8100 kg are held at a distance of 1m. The gravitational field at a point on the line joining them is zero. The gravitational potential at that point in J/kg is $\left(G = 6.67 \times 10^{-11}\ N.m^2/Kg^2\right)$

(A) $-6\ .67 \times 10^{-7}$

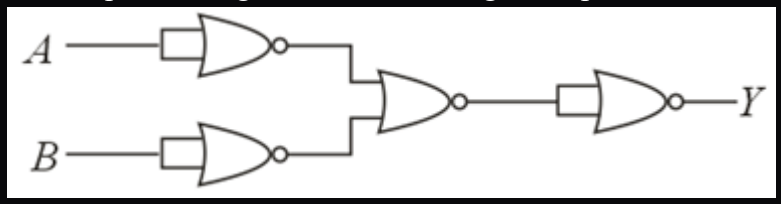
(B) -6.67×10^{-10}

(C) -13.34×10^{-7}

(D) -6.67×10^{-9}

Q.14

The output of the given combination gates represents:



(A) XOR Gate

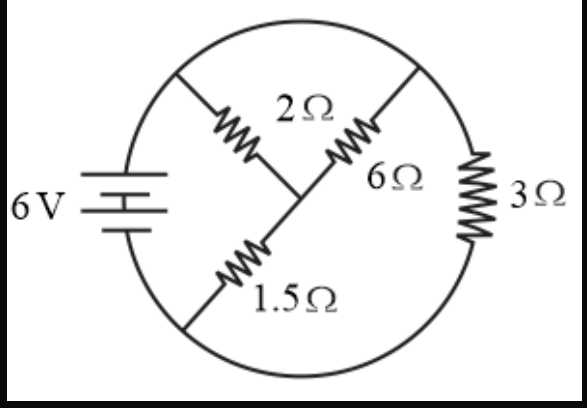
(B) NAND Gate

(C) AND Gate

(D) NOR Gate

Q.15

The total current supplied to the circuit by the battery is



(A) 1 A

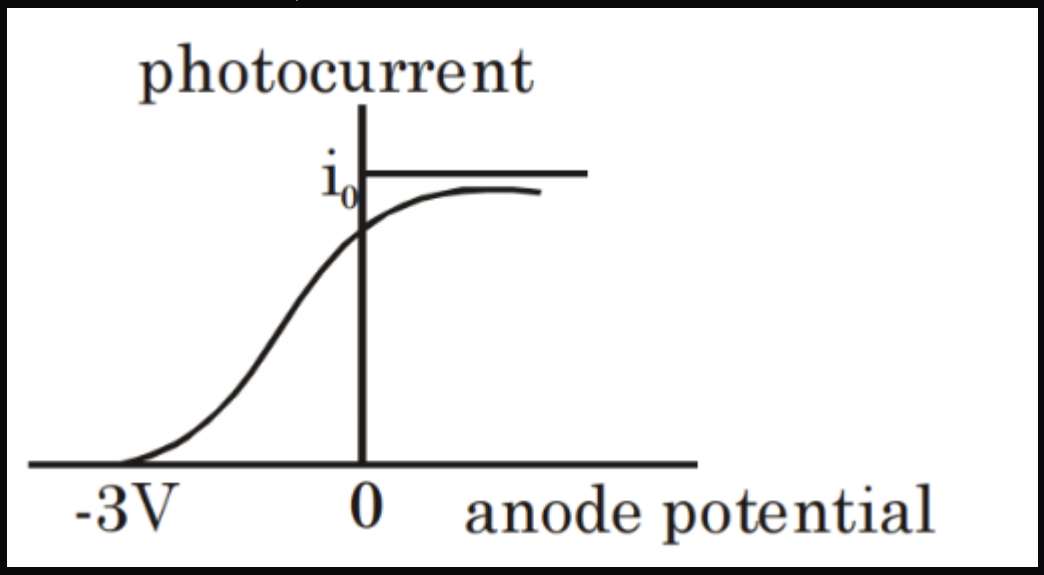
(B) 2 A

(C) 4 A

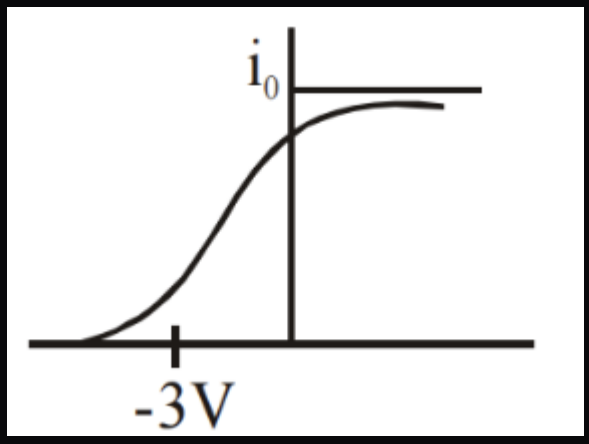
(D) 6 A

Q.16

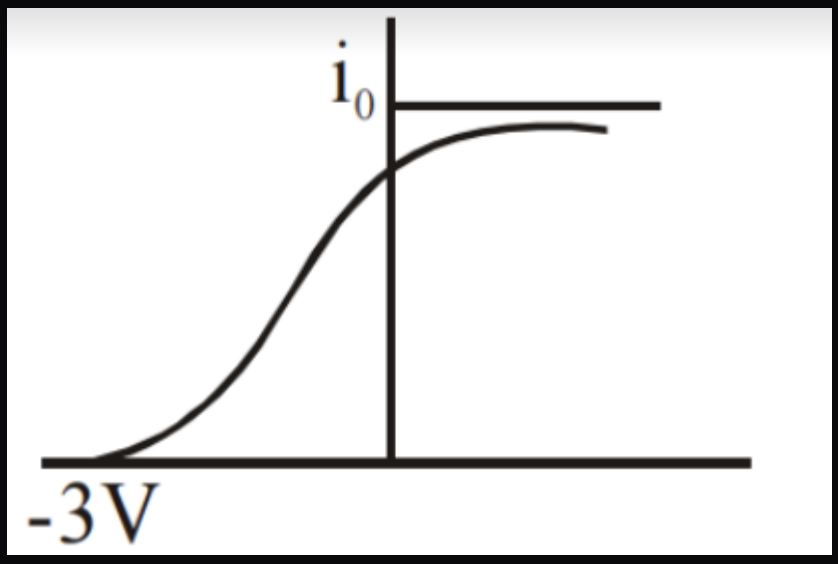
Figure shows the anode potential vs photo-current graph when photons of 6eV are incident on cathode in a photoelectric experiment set-up. In the same experimental set-up, if photons of 8eV and same intensity are incident on cathode, then anode-potential vs photo-current graph will be (assume 100% efficiency of photons to eject photo-electrons in both cases)



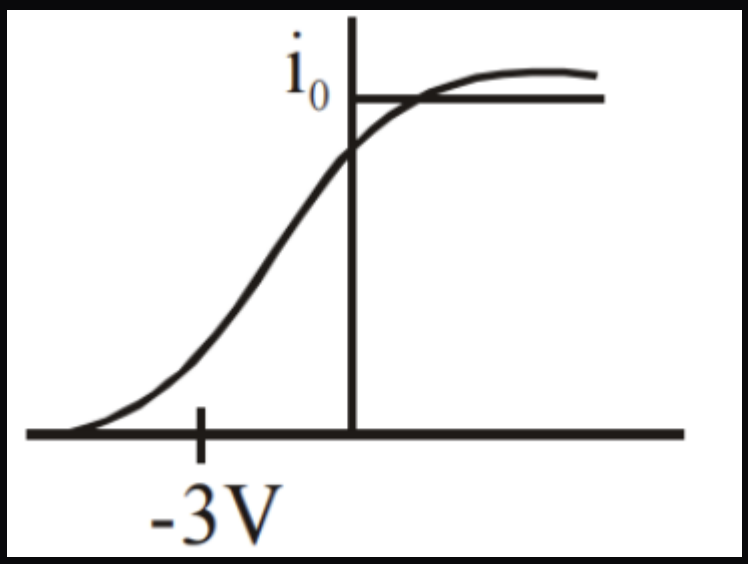
(A)



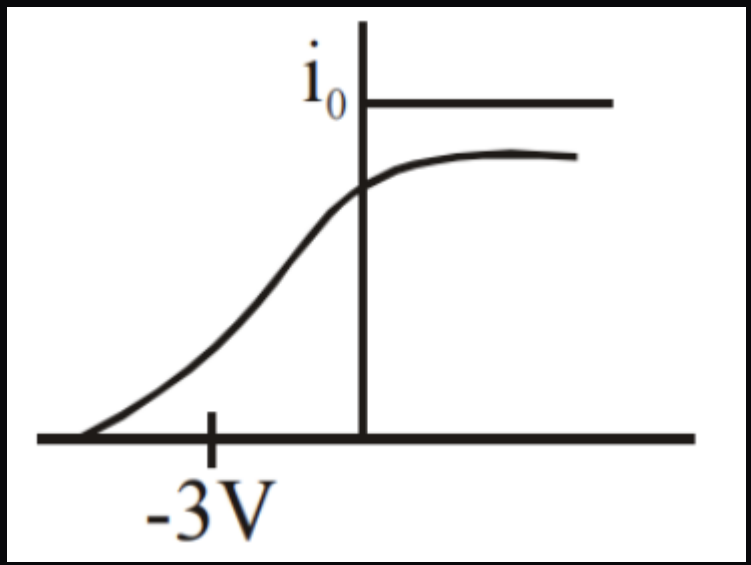
(B)



(C)



(D)



Q.17

The ratio of magnitude of magnetic field at a distance $2d$ from short bar magnet in longitudinal direction to the field at distance d in transverse direction is

(A) $1:1$

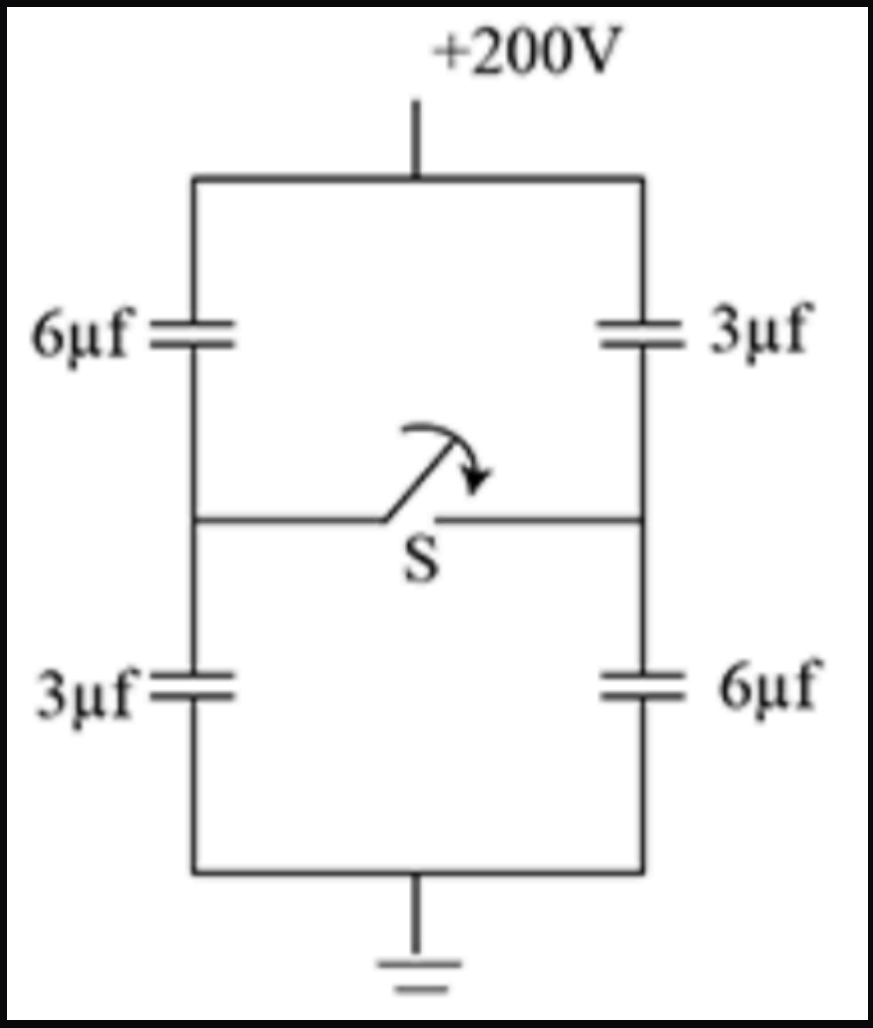
(B) $1:2$

(C) $2:1$

(D) $1:4$

Q.18

In the given circuit, charge flows through switch ' S' when it is closed.



(A) $+200\mu C$

(B) $+300\mu C$

(C) $+400\mu C$

(D) $500\mu C$

Q.19

The pressure of a fixed quantity of ideal gas is doubled. The average speed of the molecules is also doubled. The original density of the gas is ρ. Which is the new density of the gas?

(A) $\frac{\rho}{2}$

(B) ρ

(C) 2ρ

(D) 4ρ

Q.20

A rod of length l (laterally thermally insulated) of uniform cross-sectional area A consists of a material whose thermal conductivity varies with temperature as $K = \frac{k_0}{a+bT}$, where, k_0, a and b are constants. T_1 and T_2 $(< T_1)$ are the temperature of two ends of rod. Then, rate of flow of heat across the rod is

(A) $\frac{Ak_0}{bl}\left(\frac{a+bT_1}{a+bT_2}\right)$

(B) $\frac{Ak_0}{bl}\left(\frac{a+bT_2}{a+bT_1}\right)$

(C) $\frac{Ak_0}{bl}\ln\left[\frac{a+bT_1}{a+bT_2}\right]$

(D) $\frac{Ak_0}{al}\ln\left[\frac{a+bT_2}{a+bT_1}\right]$

Q.21

If one face of a prism angle 30° and $\mu = \sqrt{2}$ is silvered, the incident ray retraces its initial path. What is the angle of incidence?

Q.22

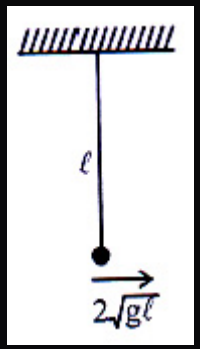
A flywheel rotates with a uniform angular acceleration. Its angular velocity increases from $20\pi \text{ rads}^{-1}$ to $40\pi \text{ rads}^{-1}$ in 10 s. How many rotations did it make in this period?

Q.23

A non-conducting ring of radius 0.5 m carries total charge of 1.11×10^{-10} C distributed non-uniformly on its circumference producting an electric field everywhere in space. The value of the line integral $\oint_{l=\infty}^{l=0} -E \cdot dl$ ($l = 0$, being centre of ring) in volt is

Q.24

One end of a string of length $\ell = \frac{14}{9}$m is fixed and a mass of 1 kg is tied to the other end. The ball is given a velocity $2\sqrt{g\ell}$ at the bottom most point as shown in figure. The string is cut when the ball becomes horizontal. Find the distance (in m) travelled till it stop for the 1st time $\left(\text{Take}\pi = \frac{22}{7}\right)$.



Q.25

A sample consisting of Hydrogen atoms in the ground state is excited by monochromatic radiation of energy

12.75eV. If we were to observe the emission spectrum of this sample, then the number of spectral lines observed, will be

Q.26

Statement - 1: Vapour above a mixture of two liquids that does not obey Raoult's law is always richer in more volatile liquid.
Statement - 2: Azeotropic mixture gives off a vapour of same composition as that of liquid.

(A) Statement-1, is True, Statement - 2 is True; Statement - 2 is a correct explanation for Statement-1.

(B) Statement - 1 is True, Statement is True; Statement - 2, is NOT a correct explanation for Statement - 1.

(C) Statement - 1 is True, Statement - 2, is False.

(D) Statement - 1 is False, Statement - 2 is True.

Q.27

It is an experimental fact that $\text{dmg} + \text{Ni(II)}$ salt $+\text{NH}_4\text{OH} \longrightarrow$ Red ppt Which of the following is wrong about this Red ppt.?

(A) It is non-ionic compound

(B) It involves intra molecular H-bonding

(C) Ni(II) is sp^3 hybridised

(D) It is a diamagnetic complex

Q.28

Aryl halides do not undergo nucleophilic substitution reactions under ordinary conditions because
(1) Approach of nucleophile is retarded
(2) Carbon carrying halogen atom is sp^3 hybridised
(3) The substrate molecule is destabilized due to resonance
(4) Partial double bond character between carbon and halogen

(A) 2 and 3 only

(B) 2, 3 and 4 only

(C) 1 and 4 only

(D) 1 and 3 only

Q.29

Match List-I with List-II. Ionization Energy is mentioned in List - II.

	List-I Electronic configuration of elements		List-II $\Delta_i H$ in kJ mol^{-1}
(a)	$1s^2 2s^2$	(p)	801
(b)	$1s^2 2s^2 2p^4$	(q)	899
(c)	$1s^2 2s^2 2p^3$	(r)	1314
(d)	$1s^2 2s^2 2p^1$	(s)	1402

Choose the most appropriate answer from the options given below:

(A) $(a) \rightarrow (q), (b) \rightarrow (r), (c) \rightarrow (s), (d) \rightarrow (p)$

(B) (a) $\rightarrow$ (p), (b) $\rightarrow$ (r), (c) $\rightarrow$ (s), (d) $\rightarrow$ (q)

(C) (a) $\rightarrow$ (p), (b) $\rightarrow$ (s), (c) $\rightarrow$ (r), (d) $\rightarrow$ (q)

(D) (a) $\rightarrow$ (s),(b) $\rightarrow$ (p), (c) $\rightarrow$ (q), (d) $\rightarrow$ (r)

Q.30

To distinguish between salicylic acid and phenol, one can use :

(A) $NaHCO_3$ solution

(B) $5\% NaOH$ solution

(C) neutral $FeCl_3$

(D) bromine water

Q.31

The increasing order of Ag^+ ion concentration in

I. Saturated solution of $AgCl$

II. Saturated solution of AgI

III. $1M Ag(NH_3)_2^+$ in $0.1M NH_3$

IV. $1M Ag(CN)_2^-$ in $0.1M KCN$

Given:

K_{sp} of $AgCl = 1.0 \times 10^{-10}$

K_{sp} of $AgI = 1.0 \times 10^{-16}$

K_d of $Ag(NH_3)_2^+ = 1.0 \times 10^{-8}$

K_d of $Ag(CN)_2^- = 1.0 \times 10^{-21}$

(A) $I < II < III < IV$

(B) $IV < III < II < I$

(C) $IV < II < III < I$

(D) $IV < II < I < III$

Q.32

The values of ΔH and ΔS for the reaction,
C (graphite) $+\mathrm{CO}_2(g) \rightarrow 2\mathrm{CO}(g)$ are $170\ \mathrm{kJ}$ and $170\mathrm{JK}^{-1}$, respectively. This reaction will be spontaneous at

(A) $910\ \mathrm{K}$

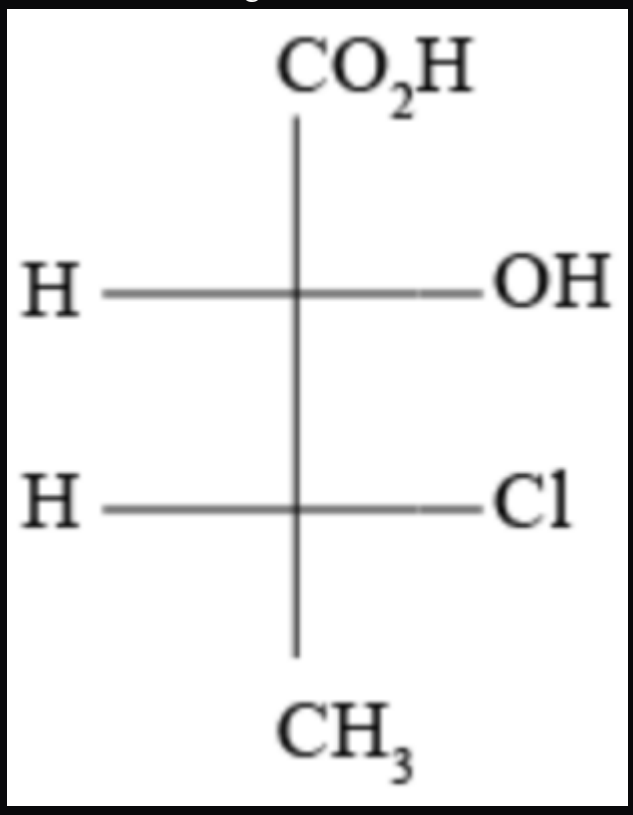
(B) $1110\ \mathrm{K}$

(C) $510\ \mathrm{K}$

(D) $710\ \mathrm{K}$

Q.33

The absolute configuration of



is

(A) $(2R, 3R)$

(B) $(2R, 3S)$

(C) $(2S, 3R)$

(D) $(2S, 3S)$

Q.34

Which of the following is correctly matched?

(A) $\mathrm{VO}_4^{3-} < \mathrm{CrO}_4^{2-} < \mathrm{MnO}_4^-$ (Oxidizing Power)

(B) $\mathrm{VO}_4^{3-} < \mathrm{CrO}_4^{2-} < \mathrm{MnO}_4^-$ (Intensity of colour)

(C) $\mathrm{CrO}_4^{2-} < \mathrm{WO}_4^{2-}$ (Stability)

(D) All are correct

Q.35

By diluting a weak electrolyte, specific conductivity $(\mathrm{K_c})$ and equivalent conductivity (λ_c) change as-

(A) Both increase

(B) K_c increases, λ_c decreases

(C) $\mathrm{K_c}$ decreases, λ_c increases

(D) Both decrease

Q.36

Which of the following statements is (are) incorrect?
(i) Percentage ionic character in $CsCl$ bond is $\sim 55\%$. (Electronegativity values of Cs and Cl are 0.7 and 3.0 respectively)
(ii) From $\mathrm{H_2O}$ to $\mathrm{H_3O^+}$, the geometry around O atom changes drastically.
(iii) The most stable oxidation state for element with atomic number 113 is expected to be $+2$.
(iv) The 2nd ionization energy of Ca is greater than 1st ionization energy of it but lower than 2nd ionization energy of K.

(A) Only i

(B) i, ii, iii

(C) ii, iii

(D) iii, iv

Q.37

Which of the following functional groups cannot be reduced to alcohol using $\mathrm{NaBH_4}$ in ethanolic solution?

(A) $R-O-R$

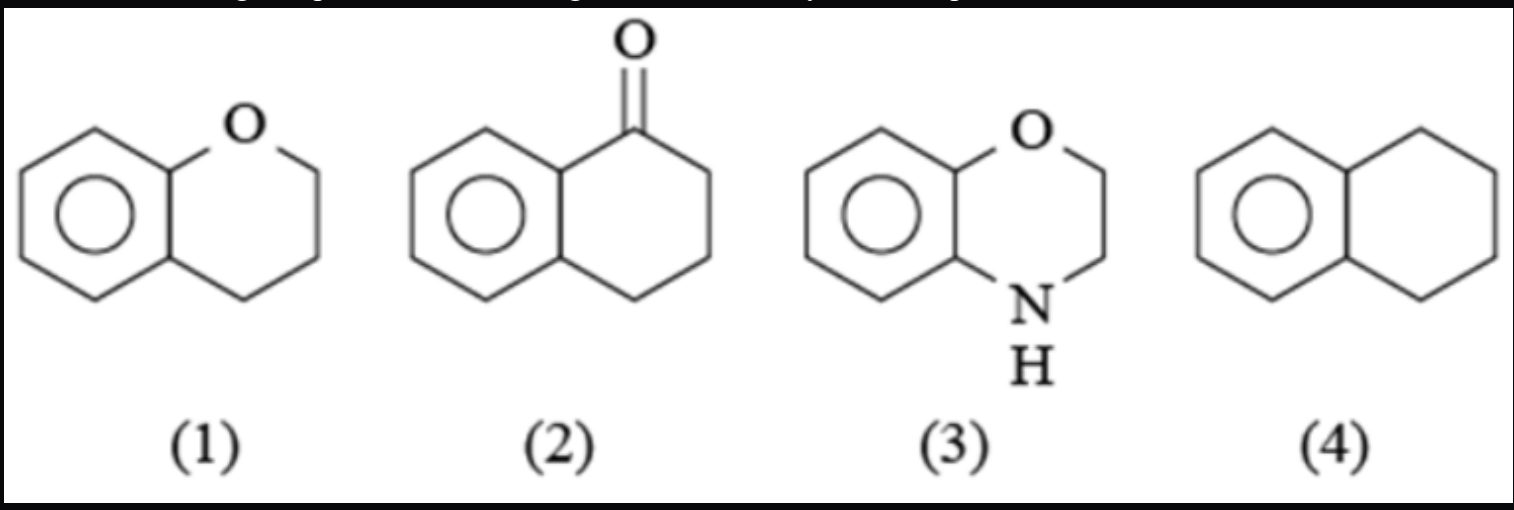
(B) RCOCl

(C) $\mathrm{R-CHO}$

(D) None of these

Q.38

Rank the following compounds in decreasing order of reactivity in electrophilic aromatic substitution reaction



(A) $3 > 1 > 2 > 4$

(B) $4 > 3 > 2 > 1$

(C) $3 > 1 > 4 > 2$

(D) $1 > 3 > 4 > 2$

Q.39

If $Ca_3\,(PO_4)_2$ and H_3PO_3 contain same number of ' P ' atom then the ratio of oxygen atoms in these compounds respectively is

(A) $8/3$

(B) $2/3$

(C) 3

(D) $4/3$

Q.40

Consider the atomization of $Br_2(g)$ according to the reaction :

$Br_2(g) \rightleftharpoons 2\,Br(g) \quad \{$ Given $e^{(-7.02)} = 9 \times 10^{-4}\}$

If the heat of atomization of bromine gas is $82\ KJ/mol$; standard molar entropies of $Br(g)$ and $Br_2(g)$ are 175 and $245.4\ JK^{-1}$ respectively, calculate degree of dissociation when the total pressure is 40 atm at $500\ K$. (assume $\alpha << 1$ in your calculation)

(A) 2.37×10^{-3}

(B) 3.20×10^{-4}

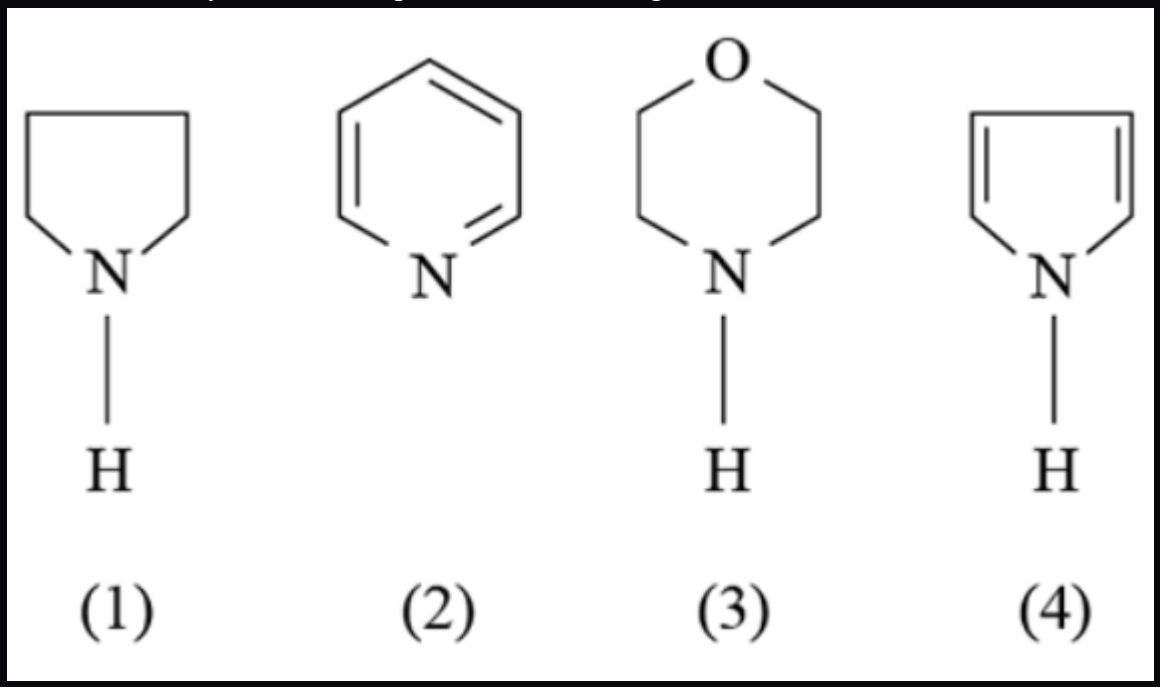
(C) 4.30×10^{-5}

(D) 3.60×10^{-2}

Q.41

Consider the following compounds

Order of basicity of these compounds in decreasing order is



(A) $4 > 1 > 2 > 3$

(B) $1 > 3 > 4 > 2$

(C) $2 > 3 > 4 > 1$

(D) $1 > 3 > 2 > 4$

Q.42

The following questions consist of two statements one labelled ASSERTION (A) and the another labelled REASON (R). Select the correct answers to these questions from the codes given below:

Assertion : Pure phosphine is non inflammable.

Reason : Impure phosphine is inflammable owing to the presence of P_2H_4 or P_4 vapours.

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true but R is not correct explanation of A

(C) A is true but R is false

(D) A and R are false

Q.43

Choose the best combination of pair of compounds that undergo aldol condensation followed by dehydration to produce but-2-enal.

(A) Methanal and ethanal

(B) Two moles of ethanal

(C) Methanal and propanone

(D) Ethanal and propanone

Q.44

What will be the nature of existence of an amino acid (containing one amino and one carboxylic acid group) in solution of $\mathrm{pH} < \mathrm{pK}_{a_1}$?

(A) It exists an anion

(B) It exists as cation

(C) It exists as zwitter ion

(D) It exists as neutral species with no charge

Q.45

Electron gain enthalpy and ionisation energy of an atom are $-a$ and $+beV$ respectively. The electronegativity of that atom on Mulliken scale is given by:

(A) $a - b$

(B) $\frac{b-a}{2}$

(C) $a + b$

(D) $\frac{a+b}{2}$

Q.46

When 5 methyl 3,3-dimethyI 4-bromoheptane is treated with KCN, it produces a product X. How many chiral centre are present in X ? Assume given IUPAC name follow naming convention of hypothetical system.

Q.47

Consider all possible isomeric ketones including stereoisomers of Molecular Weight = 100. All these isomers are independently reacted with $\mathrm{NaBH_4}$ (Note: Stereoisomers are also reacted separately). The total number of ketones that give a racemic product(s) is/are

Q.48

An exothermic reaction, $\mathrm{A} \rightarrow \mathrm{B}$, has an activation energy of $15\ \mathrm{kcalmol^{-1}}$ and the energy of reaction is $5\ \mathrm{kcalmol^{-1}}$. The activation energy in kcal $\mathrm{mol^{-1}}$ for the reaction, $\mathrm{B} \rightarrow \mathrm{A}$ is

Q.49

How many of the following are colorful gas?
I. $\mathrm{N_2O}$
II. $\mathrm{Cl_2}$
III. $\mathrm{O_3}$
IV. $\mathrm{N_2O_4}$

V. NO
VI. NO_2

Q.50

A light of wavelength $3000\tilde{A}$ falls on a metal surface. Ejected e^- is further accelerated by a potential difference of $2eV$, then final $K.E$ of the e^- is found to be 8×10^{-19} J. If threshold energy for the metal surface is ' $\phi' eV$. Then find the numerical value of 8ϕ.

(Mark answer nearest to integer)

Q.51

The marks of some students were listed out of 75. The standard deviation of marks was found to be 9. Subsequently the marks were raised to a maximum of 100 and variance of new marks was calculated. The new variance is,

(A) 144

(B) 122

(C) 81

(D) None of these

Q.52

Let $f(x) =$- $x^2 + x + p$, where p is a real number. If $g(x) = [f(x)]$ and $g(x)$ is discontinuous at $x = \frac{1}{2}$, then p cannot be (where $[.]$ represents the greatest integer function)

(A) $\frac{1}{2}$

(B) $\frac{3}{4}$

(C) $\frac{7}{4}$

(D) $-\frac{1}{4}$

Q.53

The number of positive integral solutions of the inequality $x + y + z \leq 20$ is

(A) 1008

(B) 1028

(C) 1108

(D) 1140

Q.54

Let $2\vec{a} = \vec{b} \times \vec{c} + 2\vec{b}$ where $\vec{a}$, $\vec{b}$ and $\vec{c}$ are three unit vectors, then sum of all possible values of $\left|3\vec{a} + 4\vec{b} + 5\vec{c}\right|$ is

(A) 10

(B) 12

(C) 14

(D) 16

Q.55

Let $f(x)$ be a differentiable non- decreasing function such that $\int_0^x (f(t))^3 dt = \frac{1}{x^2}\left(\int_0^x f(t)dt\right)^3 \forall x \in R - \{0\}$ and $f(1) = 1$. If $\int_0^x f(t)dt = g(x)$ then $\frac{xg'(x)}{g(x)}$ is

(A) Always equal to 1

(B) Always equal to -2

(C) May be 1 or -2

(D) Not independent of x

Q.56

The indefinite integral $I = \int \frac{\left(\sin^2 x - \cos^2 x\right)^{2019}}{(\sin x)^{2021}(\cos x)^{2021}} dx$ simplifies to (where c is an integration constant)

(A) $\frac{\left(\sin^2 x - \cos^2 x\right)^{2020}}{2020} + c$

(B) $\frac{(\tan x - \cot x)^{2020}}{2020} + c$

(C) $\frac{(\sin x - \cos x)^{2020}}{2020} + c$

(D) $\frac{\left(\tan^2 x + \cot^2 x\right)^{2020}}{2020} + c$

Q.57

The shortest distance between the lines $\frac{x-2}{2} = \frac{y-3}{2} = \frac{z-0}{1}$ and $\frac{x+4}{-1} = \frac{y-7}{8} = \frac{z-5}{4}$ lies in the interval

(A) $[0,1)$

(B) $[1,2)$

(C) $(2,3]$

(D) $(3,4]$

Q.58

If $g(x) = 2f\left(2x^3 - 3x^2\right) + f\left(6x^2 - 4x^3 - 3\right) \forall x \in R$ and $\mathrm{f''(x)} > 0, \quad \forall x \in R$, then $g'(x) > 0$ for x belonging to

(A) $\left(-\infty, -\frac{1}{2}\right) \cup (0,1)$

(B) $\left(-\frac{1}{2}, 0\right) \cup (1, \infty)$

(C) $(0, \infty)$

(D) $(-\infty, 1)$

Q.59

For a function $f(x) = \frac{2\left(x^2+1\right)}{[x]}$ (where [.] denotes the greatest integer function), if $1 \leq x < 4$, then

(A) the range of f is $\left[4, \frac{17}{2}\right)$

(B) f is bijective function

(C) the maximum value of $f(x)$ is $\frac{34}{3}$

(D) the minimum value of $f(x)$ is 4

Q.60

S and T are the foci of an ellipse and B is an end of the minor axis. If ΔSTB is equilateral, then eccentricity is

(A) $\frac{1}{4}$

(B) $\frac{1}{3}$

(C) $\frac{1}{2}$

(D) None of these

Q.61

The solution set of inequality

$\left(\tan^{-1} x\right)\left(\cot^{-1} x\right) - \left(\tan^{-1} x\right)\left(1+\frac{\pi}{2}\right) - 2\cot^{-1} x + 2\left(1+\frac{\pi}{2}\right) > \lim_{x\to-\infty}\left[\sec^{-1} x - \frac{\pi}{2}\right]$ is (where [.] denotes the greatest integer function)

(A) $(\tan 1, \tan 2)$

(B) $(-\cot 1, \cot 2)$

(C) $(-\tan 1, \tan 2)$

(D) $(-\tan 1, \infty)$

Q.62

Box A contains 2 block and 3 red balls while box B contains 3 black and 4 red balls. Out of these two boxes one is selected at random and the probability of choosing box A is double that of box B. If a red ball is drawn from the selected box, then the probability that it has come from box B is

(A) $\frac{21}{41}$

(B) $\frac{10}{31}$

(C) $\frac{12}{31}$

(D) $\frac{13}{41}$

Q.63

Let z and w be non-zero complex numbers such that $zw = |z^2|$ and $\left|z - \overline{z}\right| + \left|w + \overline{w}\right| = 4$. If w varies, then the perimeter of the locus of z is

(A) $8\sqrt{2}$ units

(B) $4\sqrt{2}$ units

(C) 8 units

(D) 4 units

Q.64

Let W denotes the words in the English dictionary. If the relation R is given by
$R = \{(x,\ y) \in W \times W :\ \text{the word } x \text{ and } y \text{ have at least one letter in common}\}$, then R is

(A) reflexive, symmetric and not transitive

(B) reflexive, symmetric and transitive

(C) reflexive, not symmetric and transitive

(D) not reflexive, symmetric and transitive

Q.65

If a, b, c are non-zero real numbers, then the minimum value of the expression $\frac{\left(a^8+4a^4+1\right)\left(b^4+3b^2+1\right)\left(c^2+2c+2\right)}{a^4b^2}$ equals

(A) 12

(B) 24

(C) 30

(D) 60

Q.66

A ray of light coming along the line $x + y - 2 = 0$ gets reflected from the line $2x - y + 1 = 0$ and goes along the line $ax + by - 12 = 0$, then

(A) $a + b = 6$

(B) $b - a = 8$

(C) $a \div b = 7$

(D) $b \times a = 7$

Q.67

The locus of the point (x, y) whose distance from the line $y = 2x + 2$ is equal to the distance from $(2, 0)$, is a parabola with the length of latus rectum same as that of the parabola $y = Kx^2$, then the value of K is equal to

(A) $\frac{\sqrt{5}}{12}$

(B) $\frac{\sqrt{5}}{4}$

(C) $\frac{4}{\sqrt{5}}$

(D) $\frac{12}{\sqrt{5}}$

Q.68

Let $A = [a_{ij}]_{3\times 3}$ be such that {{a}_{ij}}=\left\{

$$\begin{array}{ll} 3; & \text{when } \hat{i} = \hat{j} \\ 0; & \hat{i} \neq \hat{j} \end{array}$$

\right., $\left\{\frac{|Adj\,(Adj\,A)|}{5}\right\}$ equals : (where $\{\cdot\}$ denotes fractional part function)

(A) $\frac{2}{5}$

(B) $\frac{1}{5}$

(C) $\frac{2}{3}$

(D) $\frac{1}{3}$

Q.69

Let p be an odd prime number and T_P be the following set of 2×2 matrices

$$T_P = \left\{A = \begin{bmatrix} a & b \\ c & a \end{bmatrix}; a, b, c \in \{0, 1, 2, \ldots (p-1)\}\right\}$$

The number of A in T_P such that A is either symmetric or skew - symmetric or both, and $\det(A)$ is divisible by p is

(A) $(p-1)^2$

(B) $2(p-1)$

(C) $(p-2)^2$

(D) $2p-1$

Q.70

$\int_{-1}^{1} \{x^2 + x - 3\}\, dx$, where $\{x\}$ denotes the fractional part of x, is equal to

(A) $\frac{1}{3}(1 + 3\sqrt{5})$

(B) $\frac{1}{6}(1 + 3\sqrt{5})$

(C) $\frac{1}{3}(3\sqrt{5}-1)$

(D) $\frac{1}{6}(3\sqrt{5}-1)$

Q.71

Let $a_0 = 0$ and $a_n = 3a_{n-1} + 1$ for $n \geq 1$. Then the remainder obtained on dividing a_{2010} by 11 is

Q.72

Minimum integral value of k for which the equation $e^x = kx^2$ has exactly three real distinct solutions is:

Q.73

Consider a differential equation
$\left(x\tan\left(\frac{y}{x}\right) - y\sec^2\left(\frac{y}{x}\right)\right)dx + x\sec^2\left(\frac{y}{x}\right)dy = 0$
with initial condition $y(2) = \frac{\pi}{4}$, then the value of $x\tan\frac{y}{x}$ is (round off to nearest integer)

Q.74

The remainder obtained when 27^{40} is divided by 12 is

Q.75

Let $\mathrm{a}, \mathrm{b}, \mathrm{c}, \mathrm{d}, \mathrm{e}$ be five numbers satisfying the system of equations
$2a + b + c + d + e = 6$
$a + 2b + c + d + e = 12$
$a + b + 2c + d + e = 24$
$a + b + c + 2d + e = 48$
$a + b + c + d + 2e = 96,$
then $|\mathrm{c}|$ is equal to

Answers & Solutions

Q.1 Answer:

1.9 MeV

Solution:

$\Delta \mathrm{E} = (\Delta \mathrm{M})\mathrm{C}^2$
$\Delta \mathrm{M} = \left[\mathrm{m}\left(\mathrm{Be}^7\right) + \mathrm{mcn}\right) - \mathrm{m}\left(\mathrm{Li}^{+7}\right) - \mathrm{P}$
Proton have kinetic energy equal to the binding energy of process.

Q.2 Answer:

$29.4\ \mathrm{m\ s^{-1}}$

Solution:

$u_y = 0$ **and** $a_y = g = 9.8\ \text{m s}^{-2}$**.**

$s_y = u_y t + \frac{1}{2} a_y t^2$

$= 0 \times 3 + \frac{1}{2} \times 9.8 \times (3)^2$

$= 44.1\ \text{m}$**.**

Further, $v_y = u_y + a_y t = 0 + (9.8)(3)$

$29.4\ \text{m s}^{-1}$**.**

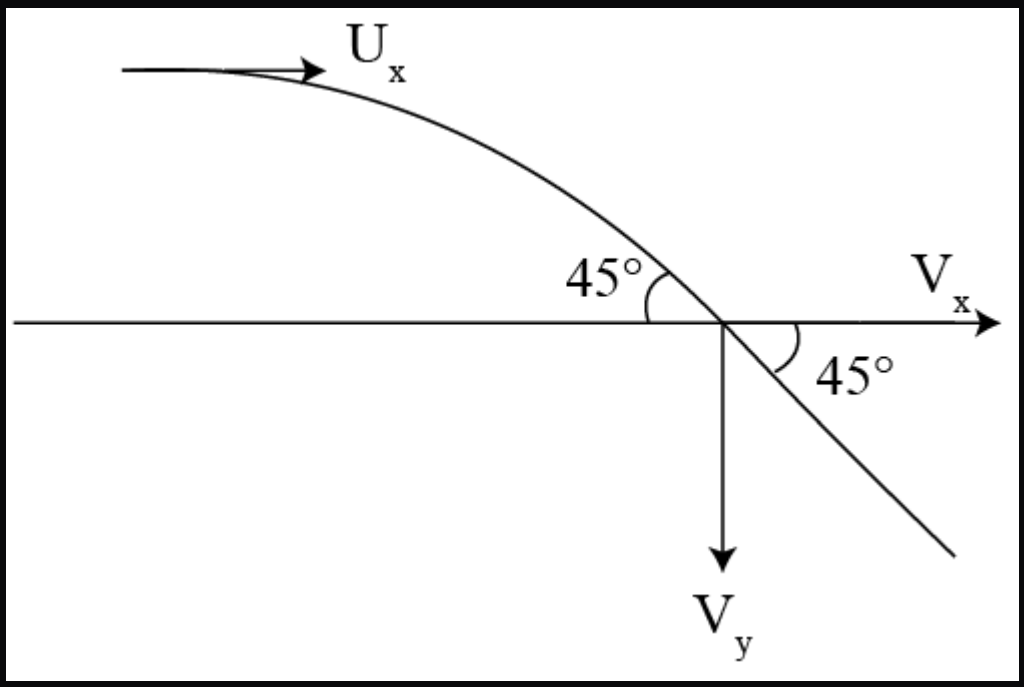
As the resultant velocity, v **makes an angle of** 45° **with the horizontal,**

$\tan 45^\circ = \frac{v_y}{v_x}$

$1 = \frac{29.4}{v_x}$

$v_x = 29.4\ \text{m s}^{-1}$**.**

Therefore, the speed with which the body was projected (horizontally) is $29.4\ \text{m s}^{-1}$**.**



Q.3 Answer:

(d) only

Solution:

Option (a) is wrong ; since in adiabatic process $V \neq$ **constant.**

Option (b) is wrong, since in isothermal process $T =$ **constant**

Option (d) matches isothermes & adiabatic formula :

$\text{TV}^{\gamma-1=}$ **constant**

$\&\frac{T^\gamma}{\text{p}^{\gamma-1}} =$ **constant**

Q.4 Answer:

$3.5\ \text{m}$

Solution:

Given

$E = 3.1\ NC^{-1} \times cos\left[(1.8\ \text{rad m}^{-1})y + (5.4 \times 10^8\ \text{rad s}^{-1})t\right]\hat{i} \quad \ldots(1)$

Comparing (1) **with the equation**

$E = E_0\ cos(ky + \omega t) \quad \ldots(2)$

We get, $k = 1.8,\ \text{rad m}^{-1}$

$E_0 = 3.1\ NC^{-1},\ c = 3 \times 10^8\ \text{ms}^{-1},\ \omega = 5.4 \times 108\ \text{rad s}^{-1}$

Now, $\lambda = \frac{2\pi}{k} = \frac{2\times22}{1.8\times7} = 3.5\ \text{m}$

Q.5 Answer:

2

Solution:

According to the question:

Time period of the simple pendulum on Earth's surface $= T_1$

Time period of the simple pendulum at a height R **above the Earth's surface** $= T_2$

$R =$ **Radius of the Earth**

Now, the time period of a simple pendulum (T)**,**

$T = 2\pi\sqrt{\frac{l}{g}}$

Here, $l =$ **Length of the simple pendulum**

$g =$ **Accceleration due to gravity on Earth's surface.**

As in both the cases, length of the simple pendulum remains same, so,

$T \propto \frac{1}{\sqrt{g}}$

Relation of variation in g **due to height** (h) **above the Earth's surface** (g_h)

$g_h = \frac{g}{\left(1+\frac{h}{R}\right)^2}$

Accceleration due to gravity at a height $(h = R)$**,**

$g_R = \frac{g}{\left(1+\frac{R}{R}\right)^2} \Rightarrow \frac{g}{(1+1)^2}$

$g_R = \frac{g}{4}$

From $T \propto \frac{1}{\sqrt{g}}$

$T_1 \propto \frac{1}{\sqrt{g}}$

$T_2 \propto \frac{1}{\sqrt{g_R}}$

Take ratio of $T_2\ :\ T_1$

$\frac{T_2}{T_1} = \sqrt{\frac{g}{g_R}}$

Put the value of g_R **in the above relation,**

$\frac{T_2}{T_1} = \sqrt{\frac{g}{\frac{g}{2}}} \Rightarrow \sqrt{4}$

$\frac{T_2}{T_1} = 2$

Q.6 Answer:

$ml^2/2\pi^2$

Solution:

Here, $\pi r = l\ \ \therefore r = l/\pi$

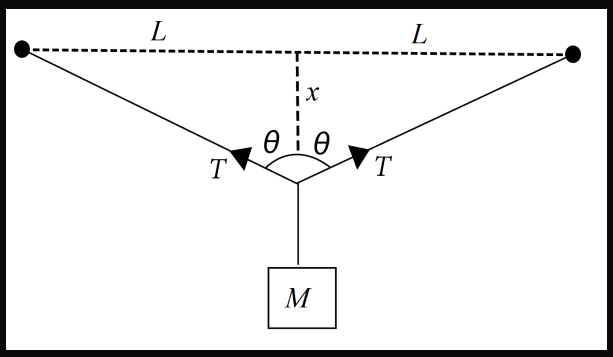
Moment of inertia of a ring about its diameter $= \frac{1}{2}\text{Mr}^2$

$\therefore$ **Moment of inertia of semicircle** $= \frac{1}{2}\left[m\left(\frac{l}{\pi}\right)^2\right] = \frac{ml^2}{2\pi^2}$

Q.7 Answer:

5.92

Solution:



Change in length,

$$\Delta L = 2\left(\sqrt{\mathrm{L}^2+\mathrm{x}^2}-\mathrm{L}\right) = 2\left[\mathrm{L}\left(1+\frac{\mathrm{x}^2}{\mathrm{L}^2}\right)^{1/2}-\mathrm{L}\right] \Rightarrow \Delta L = 2\left[\mathrm{L}\left(1+\frac{\mathrm{x}^2}{2\mathrm{L}^2}\right)-\mathrm{L}\right] = \frac{\mathrm{x}^2}{\mathrm{L}}$$

At equilibrium

$$2\mathrm{T}\cos\theta = \mathrm{mg}$$

$$\Longrightarrow 2\left(\frac{\mathrm{YA}\Delta\mathrm{L}}{2L}\right)\cos\theta = \mathrm{mg} \Longrightarrow \left(\frac{\mathrm{YA}\,x^2}{\mathrm{L}^2}\right)\times\frac{x}{L} = \mathrm{mg}$$

$$\Rightarrow \frac{\mathrm{YAx}^3}{\mathrm{L}^3} = \mathrm{mg}$$

$$\Rightarrow \mathrm{x} = \left\{\frac{\mathrm{L}^3\mathrm{mg}}{\mathrm{YA}}\right\}^{\frac{1}{3}}$$

$$\Rightarrow \mathrm{x} = \left\{\frac{1\times5\times10}{2.4x10^9\times10^{-4})}\right\}^{\frac{1}{3}}$$

$$\Rightarrow x = 5.92\ \mathrm{cm}$$

Q.8 Answer:

A cricket ball moving with a uniform speed hit with a bat for a very short time interval

Solution:

The $a-\mathrm{t}$ graph shows that initially the body moves with uniform velocity. Its acceleration increases for a short time and then falls to zero and there after the body moves with a constant velocity. Such a physical Situation arises when a cricket ball moving with a uniform speed is hit with a bat for a very short time interval.

Q.9 Answer:

Assertion is False but Reason is True.

Solution:

The equation of displacement in linear simple harmonic motion is $y = A\sin\omega t$

Velocity is the rate of change of displacement

$$v = \frac{\mathrm{d}y}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}A\sin\omega t = \omega A\cos\omega t = \omega A\sin\left(\omega t+\frac{\pi}{2}\right)$$

Q.10 Answer:

$0.59\ \mathrm{cm}$

Solution:

L.C. of Vernier calipers

$= \frac{1 \, main \; scale \, division}{Total \; divisons \; \text{vernier} \; scale}$

$= \frac{0.1}{10} = 0.01 \ \text{cm}$

Reading of Vernier calipers

$= M.S.R. \ + (L.C) \times vs \; divisions.$

$d_1 = 0.5 + 0.01 \times 8$

$d_2 = 0.5 + 0.01 \times 4$

$d_3 = 0.5 + 0.01 \times 6$

$\therefore$ **Measured diameter are respectively**

$0 \ .58 \ \ \text{cm} \quad 0 \ .54 \ \ \text{cm}, \quad 0 \ .56 \ \ \text{cm}$

$\therefore \quad average \; diameter \; = \; \frac{0.58 + 0.54 + 0.56}{3}$

$= \frac{1.68}{3} = 0.56$

Negative zero error is added in the average diameter to get the corrected diameter

$\therefore \quad corrected \; diameter = 0.56 - (-0.03)$

$= 0.56 + 0.03 = 0.59 \ \text{cm}$

Q.11 Answer:

16%

Solution:

$\frac{\Delta X}{X} \times 100 = 2\left(\frac{\Delta A}{A} \times 100\right) + \frac{1}{2}\left(\frac{\Delta B}{B} \times 100\right) + \frac{1}{3}\left(\frac{\Delta C}{C} \times 100\right) + 3\left(\frac{\Delta D}{D} \times 100\right)$

$= 2 \times 1 + \frac{1}{2} \times 2 + \frac{1}{3} \times 3 + 3 \times 4$

$= 16\%$

Q.12 Answer:

If Assertion is false but Reason is true.

Solution:

Coherent sources can be created from same source. Different sources are always incoherent.

Q.13 Answer:

$-6 \ .67 \times 10^{-7}$

Solution:

Let us assume the point at which gravitational field is zero is at x distance from $100 \ \text{kg}$

So magnitude of gravitational field must be same at that point due to both masses, so

$\Rightarrow \frac{G \times 100}{x^2} = \frac{G \times 8100}{(1-x)^2}$

$\Rightarrow \frac{100}{x^2} = \frac{8100}{(1-x)^2}$

$\Rightarrow \frac{10}{x} = \frac{90}{(1-x)}$

$\Rightarrow 10(1-x) = 90x$

$\Rightarrow 10 - 10x = 90x$

$\Rightarrow 100x = 10$

$\Rightarrow x = 0.1\ \text{m}$

Now gravitational potential at that point due to both masses is given by,

$\Rightarrow V = -\frac{G\times 100}{x} - \frac{G\times 8100}{(1-x)}$

$\Rightarrow V = -G \times 100\left(\frac{1}{0.1} + \frac{81}{(1-0.1)}\right)$

$\Rightarrow V = -6.67 \times 10^{-11} \times 100\left(10 + \frac{81}{0.9}\right)$

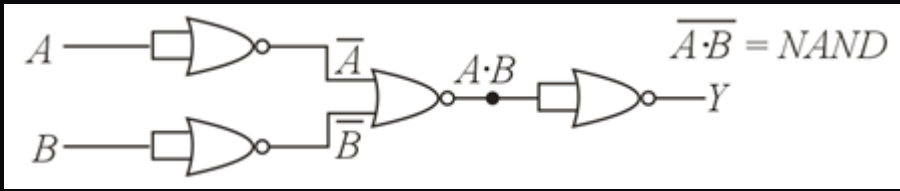
$\Rightarrow V = -6.67 \times 10^{-11} \times 100(10+90) = -6.67 \times 10^{-7}\ \text{J kg}^{-1}$

Q.14 Answer:

NAND Gate

Solution:

By De Morgan's theorem, we have

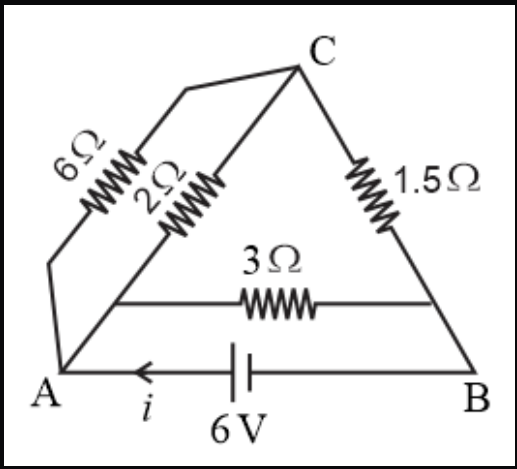


Q.15 Answer:

4 A

Solution:

Let us assume that the battery of $6\ \text{V}$ is connected across points A and B. Let the centre point be C. Using the point-potential method, the points having the same potential should have the same name. After redrawing the circuit, the figure looks as shown below,



The current supplied by the battery is

$i = \frac{V}{R_{\text{AB}}}$

$V = 6\ \text{V}$.

Now, to find the equivalent resistance between A and B,

$R_{\text{AB}} = 3 \parallel (R_{\text{AC}} + R_{\text{CB}})$

$R_{\text{AC}} = \frac{6\times 2}{6+2}$

$R_{\text{AC}} = 1.5\ \Omega$

$R_{\text{BC}} = 1.5\ \Omega$

$R_{AB} = 3 \parallel (1.5 + 1.5)$

$R_{AB} = 3 \parallel 3$

$R_{AB} = \frac{3 \times 3}{3+3}$

$R_{AB} = 1.5\ \Omega.$

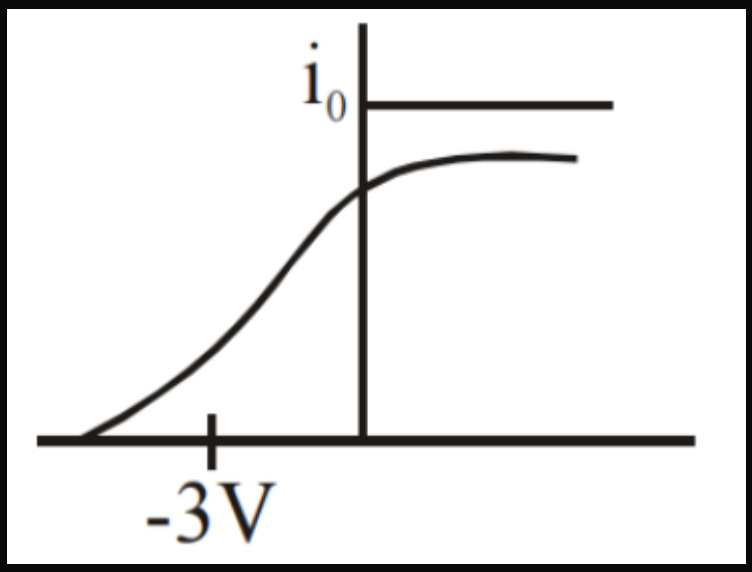
Thus, the net resistance between A and B is $1.5\ \Omega$.

Now, the current flowing through the battery is

$i = \frac{6}{1.5}$ A.

Hence, the current flowing through the battery is 4 A.

Q.16 Answer:



Solution:

Stopping potential increases and number of photons decreases.

$\mathrm{I} = \frac{\mathrm{hcn_p}}{\lambda \mathrm{t}}$

Q.17 Answer:

$1 : 4$

Solution:

For longitudinal direction, $B = \frac{\mu_0}{4\pi}\frac{2M}{r^3}$

So, for given case $B_l = \frac{\mu_0}{4\pi} \times \frac{2M}{(2d)^3} = \frac{\mu_0 M}{4\pi \times (4d^3)}$

For transverse direction, $B_t = \frac{\mu_0}{4\pi} \times \frac{M}{r^3}$

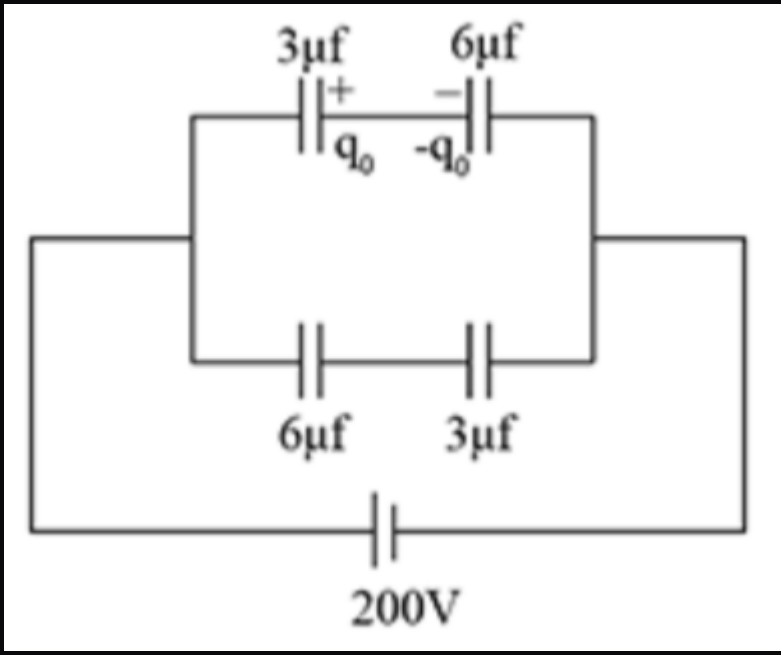
$\Rightarrow = \frac{\mu_0}{4\pi} \times \frac{M}{d^3}$

Hence, $\frac{B_I}{B_t} = \frac{1}{4}$

Q.18 Answer:

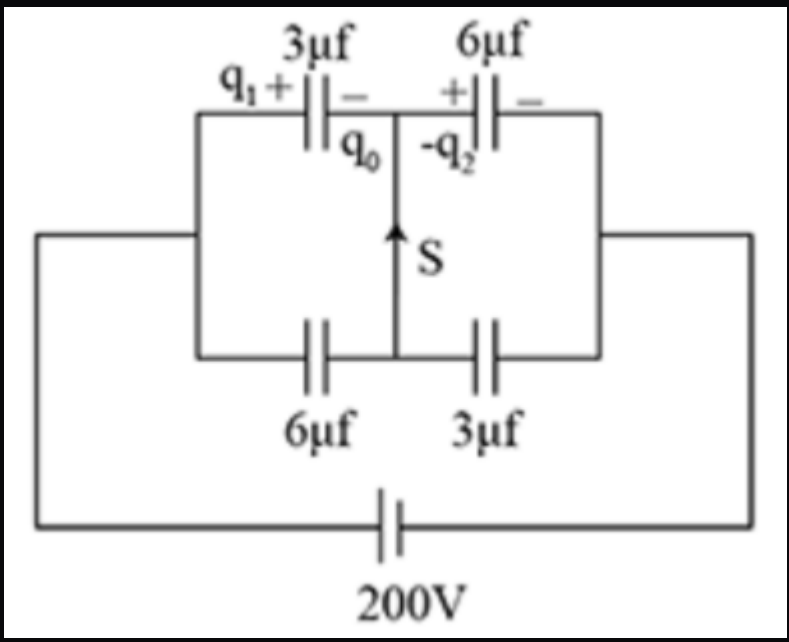
$+300\mu C$

Solution:



When switch is opened $q_0 = Ceq \times \varepsilon = 400\mu C$

When switch is closed



$q_1 = 300\mu C, \quad q_2 = 600\mu C$

Q.19 Answer:

$\frac{\rho}{2}$

Solution:

From the kinetic theory of gas we get to know that the pressure of a gas is:

$P = \frac{1}{3}\rho c^2$**. Here** ρ **is the density of the gas and** c **is the average speed of the gas molecules.**

$\therefore \ \rho = \frac{3P}{c^2}$**.**

Now, if pressure and average speed both are doubled, then the new density becomes:

$\rho_{new} = \frac{3\times 2P}{(2c)^2} = \frac{1}{2} \times \left(\frac{3P}{c^2}\right) = \frac{\rho}{2}$**.**

$\therefore$ **The new density of the gas is** $\frac{\rho}{2}$**.**

Q.20 Answer:

$\frac{Ak_0}{bl} \ln \left[\frac{a+bT_1}{a+bT_2}\right]$

Solution:

$$\text{As, } \frac{dQ}{dt} = -KA\frac{dT}{dx}$$
$$\Rightarrow \frac{dQ}{dt} = -\frac{k_0 A}{a+bT}\frac{dT}{dx}$$
$$\Rightarrow \frac{dQ}{dt} = -\frac{k_0 A}{a+bT}\frac{dT}{dx}$$

On integrating both sides within the proper limits, we get $\frac{dQ}{dt}\int_0^1 dx = -k_0 A \int_{T_1}^{T_2} \frac{dT}{a+bT}$

This gives $\frac{dQ}{dt} = \frac{Ak_0}{bl} \ln \left[\frac{a+bT_1}{a+bT_2}\right]$

Q.21 Answer:

—

Solution:

As incident ray retraces its path the ray is incident normally on the silvere face of the prism as shown in figure. Further, as in

$\triangle \text{AED} \quad 30 + 90 + \angle \text{D} = 180$

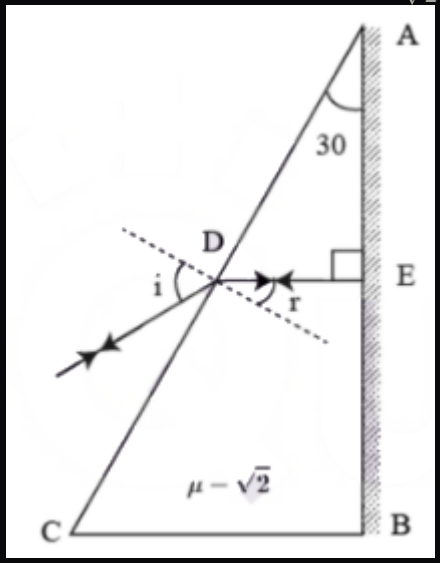
$\Rightarrow \angle D = 60$

Now as by construction, $\angle \text{D} + \angle \text{r} = 90 \Rightarrow$

$\angle \text{r} = 90 - 60 = 30$

∴ from Snell's law at surface $\text{AC}, 1 \sin \text{i} =$

$\sqrt{2} \sin 30^\circ = \sqrt{2} \times \frac{1}{2} = \frac{1}{\sqrt{2}} \quad \therefore \sin \text{i} = \frac{1}{\sqrt{2}} \Rightarrow \text{i} = 45^\circ$



Q.22 Answer:

—

Solution:

As, $\omega_2 = \omega_1 + \alpha t$

$\therefore \quad 40\pi = 20\pi + \alpha \times 10$

or$\alpha = 2\pi \text{rads}^{-2}$

From,$\omega_2^2 - \omega_1^2 = 2\alpha\theta$

$(40\pi)^2 - (20\pi)^2 = 2 \times 2\pi\theta$

$\Rightarrow \quad \theta = \frac{1200\pi^2}{4\pi} = 300\pi$

Number of rotations completed $= \frac{\theta}{2\pi} = \frac{300}{2\pi} = 150$

Q.23 Answer:

—

Solution:

From definition of potential difference

$\int_{i=\infty}^{i=0} \mathrm{E} \cdot \mathrm{dl}$ = potential difference at infinity and at centre of ring $= \mathrm{V}_{\text{centre}} -$

$\mathrm{V}_{\text{infinity}}$

But by convention $\mathrm{V}_{\text{infinity}} = 0$ and $\mathrm{V}_{\text{centre}} = \frac{1}{4\pi\varepsilon_0}\frac{\mathrm{q}}{\mathrm{R}}$

$$= 9 \times 10^9 \times \frac{1.11 \text{ times } 10^{-10}}{0.5} = 2 \text{ volt}$$

$$\therefore \int_{i=\infty}^{i=0} (-\mathrm{E} \cdot \mathrm{dl}) = 2\text{volt} - 0 \text{ volt } = 2 \text{ volt.}$$

Q.24 Answer:

—

Solution:

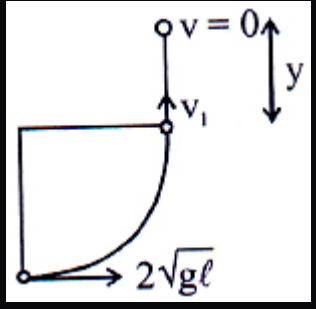
By energy conservation

$\frac{1}{2}m\left(2\sqrt{gl}\right)^2 = mgh$

$h = 2l$

$y = l$

$\text{distance} = \frac{\pi\ell}{2} + \ell = \frac{14}{9}\left[\frac{11}{7} + 1\right] = 4\text{m}$



Q.25 Answer:

—

Solution:

The energy of an electron of a hydrogen atom in the ground state is $E_0 = -13.6\text{eV}$
The energy of incoming photon $= 12.75\text{eV}$
The energy of an electron in the excited state is

$$-\frac{13.6}{n^2} = -13.6 + 12.75 = -0.85\text{eV}$$

$$\Rightarrow n^2 = \frac{13.6}{0.85} = 16$$

$$\Rightarrow n = 4$$

No. of spectral lines $= \frac{n(n-1)}{2} = 6$

Q.26 Answer:

Statement - 1 is False, Statement - 2 is True.

Solution:

Azeotropes are constant boiling mixtures. composition of components remains same in vapour phase as well as in liquid phase in azeotropic mixture.
Vapour above a mixture of two liquids that does not obey Raoult's law is always richer in more volatile liquid it is incorrect for azeotropic mixture.

Q.27 Answer:

Ni(II) is sp^3 hybridised

Solution:

Red ppt is $[\text{Ni(dmg)}_2]$
Here Ni is in $+2$ oxidation state
It is diamagnetic and dsp ^{2} hybridized

Q.28 Answer:

1 and 4 only

Solution:

Aryl halides do not undergo nucleophilic substitution reactions under ordinary conditions because approach of nucleophile is retarded and it is also due to partial double bond character between carbon and halogen which is resonance stabilized so removal of halide ions gets tough as more energy is needed to break C-X bond.

Q.29 Answer:

$(\text{a}) \rightarrow (\text{q}), (\text{b}) \rightarrow (\text{r}), (\text{c}) \rightarrow (\text{s}), (\text{d}) \rightarrow (\text{p})$

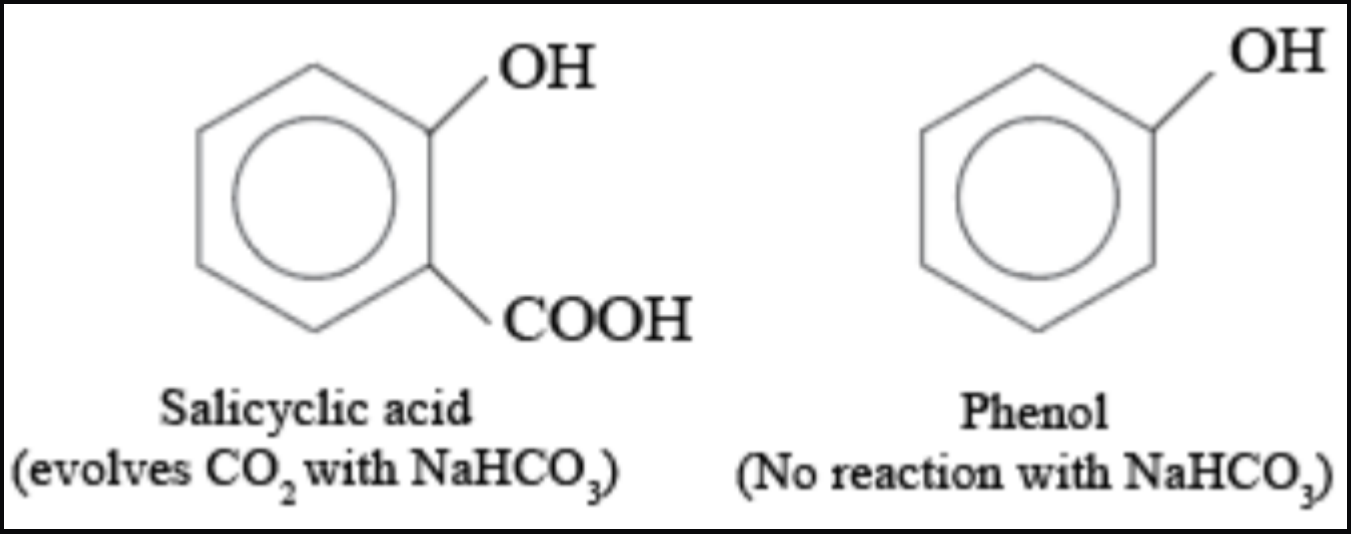
Solution:

(i) Electron affinity of second period p-block element is less than third period p-block element due to small size of second period p-block element, due to small size electron density is more and upcoming electron face extra repulsion, so energy release is less. Hence, EA order: $\mathrm{F} < \mathrm{Cl}$.
(ii) Down the group electron affinity decreases due to size increases. EA order: $\mathrm{S} > \mathrm{Se}$ $\mathrm{Li} > \mathrm{Na}$.
Ionization energy is reverse of electron afinity

Q.30 Answer:

$NaHCO_3$ solution

Solution:



Q.31 Answer:

$IV < II < III < I$

Solution:

$[\mathrm{I}]\ \left[\mathrm{Ag}^{+}\right]=\sqrt{\mathrm{K}_{\mathrm{sp}}(\mathrm{AgCl})}=\sqrt{1\times10^{-10}}=10^{-5}\mathrm{M}$
$[\mathrm{II}]\ \left[\mathrm{Ag}^{+}\right]=\sqrt{\mathrm{K}_{\mathrm{sp}}(\mathrm{AgI})}=\sqrt{1\times10^{-16}}=10^{-8}\mathrm{M}$
$[\mathrm{III}]\ \left[\mathrm{Ag}\left(\mathrm{NH}_3\right)_2^{+}\right]\rightleftharpoons\mathrm{Ag}^{+}+2\mathrm{NH}_3\ \ \mathrm{K}_{\mathrm{d}}=1\times10^{-8}=\frac{\left[\mathrm{Ag}^{+}\right]\left[\mathrm{NH}_3\right]^2}{\mathrm{Ag}[\mathrm{NH}_3]_{2\mathrm{sq}}^{+}}=\frac{\left[\mathrm{Ag}^{+}\right]0.1^2}{1.0}\Rightarrow\left[\mathrm{Ag}^{+}\right]=1\times10^{-6}\mathrm{M}$
$[\mathrm{IV}]\quad \mathrm{Ag(CN)}_{2_{(\mathrm{aq})}-1}\rightleftharpoons\mathrm{Ag}_{(\mathrm{aq})}^{+}+2\mathrm{CN}_{(\mathrm{aq}}^{-1}$
$\mathrm{KCN}(0.1\mathrm{M})\longrightarrow\mathrm{K}_{\mathrm{aq}}^{+}+\mathrm{CN}_{\mathrm{aq}}^{-1}$
$\mathrm{K}_{\mathrm{d}}=1\times10^{-21}=\frac{\left[\mathrm{Ag}^{+}\right]\left[\mathrm{CN}^{-}\right]^2}{\left[\mathrm{Ag(CN)_2}\right]^{-1}}=\frac{\left[\mathrm{Ag}^{+}\right][0.1]^2}{1.0}$
$\left[\mathrm{Ag}^{+}\right]=1\times10^{-19}\mathrm{M}$

Q.32 Answer:

1110 K

Solution:

For a spontaneous reaction, $\triangle\mathrm{G}=$ **Negative**
And at equilibrium $\triangle\mathrm{G}=0$ **i.e** $\triangle\mathrm{H}=\mathrm{T}\triangle\mathrm{S}$

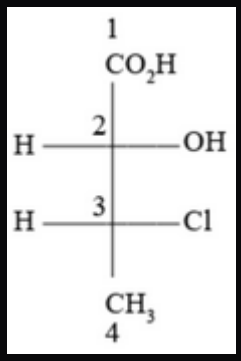
$$\Rightarrow \mathrm{T} = \frac{170 \times 10^3}{170}$$
$$\Rightarrow \mathrm{T} = 1000\ \mathrm{K}$$

Therefore, at this temperature, reaction is at equilibrium i.e $\Delta \mathrm{G} = 0$ and above this temperature $\Delta \mathrm{G}$ will be negative and reaction will be spontaneous.

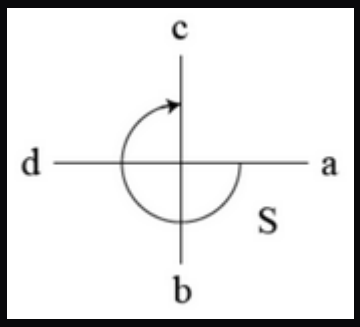
Q.33 Answer:

$(2S, 3R)$

Solution:

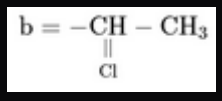


For 2nd carbon



the priority order $a > b > c > d$

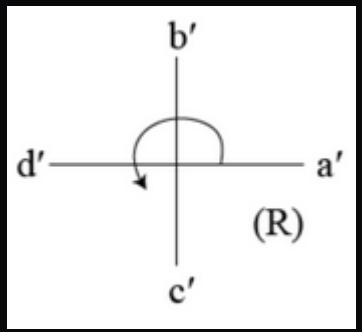
$\mathrm{a} = -\mathrm{OH}$



$\mathrm{c} = -\mathrm{CO_2H}$

$\mathrm{d} = -\mathrm{H}$

For 3rd carbon



the priority order $\mathrm{a}' > \mathrm{b}' > \mathrm{c}' > \mathrm{d}'$

$a' = -Cl$
$b' = -\underset{OH}{CH} - CO_2H$
$c' = -CH_3$
$d' = -H$

Q.34 Answer:

All are correct

Solution:

(a) $VO_4^{3-} < CrO_4^{2-} < MnO_4^- \longrightarrow$ Oxidizing power.
This is due to increasing stability of the lower oxidation state to which they are reduced.
(b) Due to charge transfer.
(c) In d-Block elements, as we proceed down a group, Stability of higher oxidation state increases.

Q.35 Answer:

K_c decreases, λ_c increases

Solution:

When the solution of a weak electrolyte is diluted, the volume of the solution increases, hence equivalent conductivity (λ_c) increases. However, during this process, the number of current carrying particles per cm^2 decreases, hence specific conductivity (K_c) decreases.

Q.36 Answer:

ii, iii

Solution:

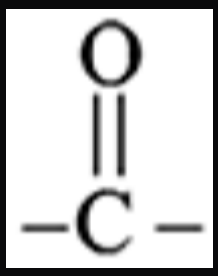
(a) Ionic character
$\%$ ionic character $= 16\,(x_A - x_B) + 3.5\,(x_A - x_B)^2$
where, $x_A - x_B =$ Difference in electro-negativity of bonded atoms.
$16 \times (3.0 - 0.7) + 3.5 \times (3.0 - 0.7)^2 = 55.32\%$
(b) Geometry for both the species is tetrahedral, as in both ' O ' is sp^3 hybridised
(c) $113 = [Rn]5f^{14}6d^{10}7s^27p^1 =$ Due to inert pair effect $7s^2$ electron do not take part in bond formation, so, most stable oxidation state is $+1$.
(d) Successive ionization energies are higher for every element. 2nd ionization of K happens from $3s^23p^6$ configuration which is an inert gas configuration and for Ca it happens from $4s^1$. So, K has higher second ionisation energy than Ca.

Q.37 Answer:

$R - O - R$

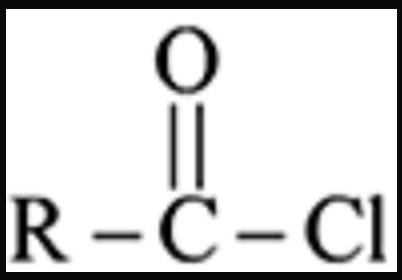
Solution:

Compounds having

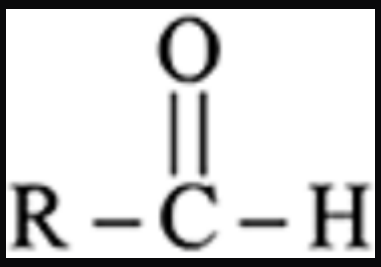


moiety in their structure are reduced to alcohols using $\mathrm{NaBH_4}$ in ethanolic solution.

Thus



and



reduced while, $\mathrm{R-O-R}$ (ethers) and carboxylic acids are inert and can't be reduced by $\mathrm{NaBH_4}$.

Q.38 Answer:

$3 > 1 > 4 > 2$

Solution:

Rate of electrophilic aromatic substitution is effected by group/s already present. If they $(-\mathrm{O}-, -\mathrm{NH}-, -\mathrm{CH_2}-)$ are ring activating and ortho, para directing due to $+\mathrm{R}$ and $+\mathrm{H}$ effects reactivity is more but if they $(-\mathrm{CO}-, -\mathrm{NO_2}, -\mathrm{CN})$ are ring deactivating and meta directing groups due to $-\mathrm{R}$ and $-\mathrm{H}$ effects reactivity decreases.

Here reactivity order is $3 > 1 > 4 > 2$.

Q.39 Answer:

$4/3$

Solution:

$$\mathrm{Ca_3\,(PO_4)_2} = \mathrm{xmol} \begin{matrix} \nearrow & 2\mathrm{xmol'P'\ atoms} \\ \searrow & 8\mathrm{xmol'0'\ atoms} \end{matrix}$$

$$\mathrm{H_3PO_3} = \mathrm{ymol} \begin{matrix} \nearrow & \mathrm{ymol'P'\ atoms} \\ \searrow & 3\mathrm{ymol'0'\ atoms} \end{matrix}$$

$2x = y$(Same 'P' atoms given)$x = \frac{y}{2}$

Ratio of '0' atom $= \frac{8x}{3y} = \frac{8 \times \left(\frac{y}{2}\right)}{3y} = \frac{4}{3}$

Q.40 Answer:

2.37×10^{-3}

Solution:

For $Br_2(g) \rightleftharpoons 2Br(g)$

$\Delta S^{\circ}_{reac} = 2 \times 175 - 244.4 = 105.6 \text{ J K}^{-1}$

$\Delta H^{\circ}_{reac} = 82 KJm^{-1}$

$\Delta G^{\circ}_{reac} = \Delta H^{\circ}_{reac} - T\Delta S^{\circ} = 82000 - 500 \times 105.6$

$= 29.2 \times 10^3 \text{ J} = -RT \ln K_P$

$\ln K_P = \frac{-29700}{8.314 \times 500} = -7.02$

$\therefore K_P = 9 \times 10^{-4}$

$K_P = \frac{4\alpha}{1-\alpha^2}$

as $\alpha << 1$

$\therefore K_P = 4 \times \alpha^2 P$

$\because P = 40$

$\therefore 9 \times 10^{-4} = 4 \cdot \alpha^2 \cdot 40$

$\therefore \alpha^2 = \frac{9}{160} \times 10^{-4}$

$= \frac{9}{16} \times 10^{-5}$

$= 2.37 \times 10^{-3}$

Q.41 Answer:

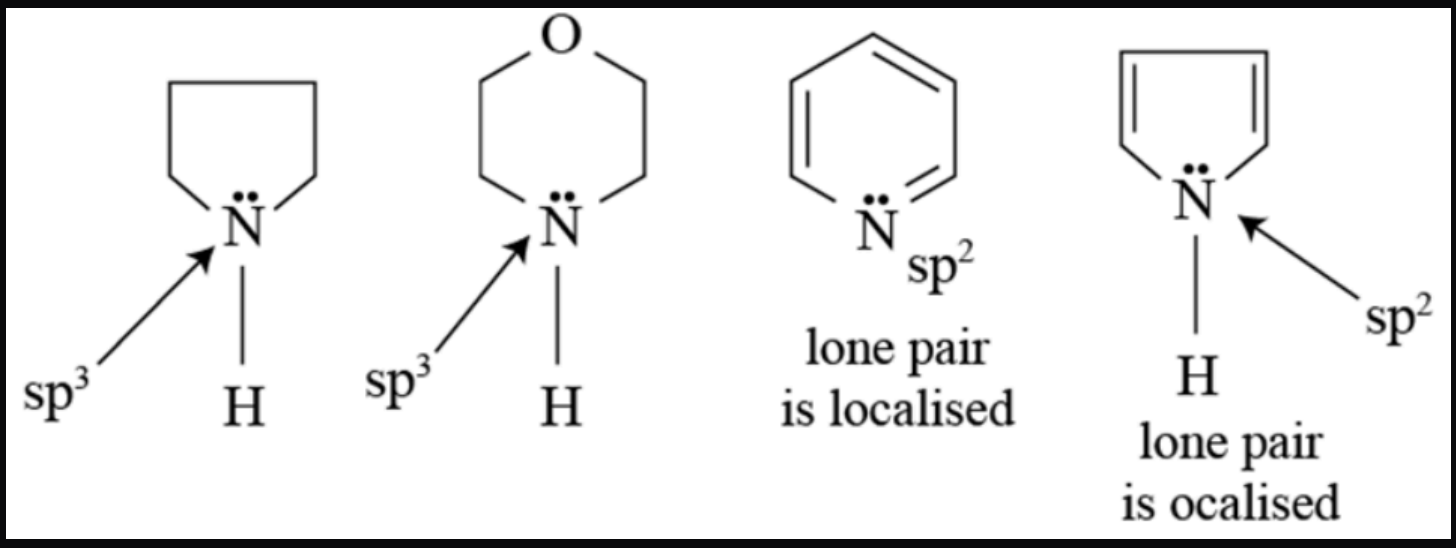
$1 > 3 > 2 > 4$

Solution:

(i) Basicity $\propto \frac{1}{\text{Electronegativity of nitrogen}}$

(ii) -I group decreases basicity

(iii) Delocalisation decreases basicity.



Q.42 Answer:

Both A and R are true and R is the correct explanation of A.

Solution:

Phosphine is a colourless, inflammable and toxic gas having molecular formula PH_3. Pure phosphine is colourless and non-inflammable, but due to the mixing of P_2H_4 or P_4 vapours it has an extremely pungent smell and becomes burns with a luminous flame.
Hence, both A and R are true and R is the correct explanation of A.

Q.43 Answer:

Two moles of ethanal

Solution:

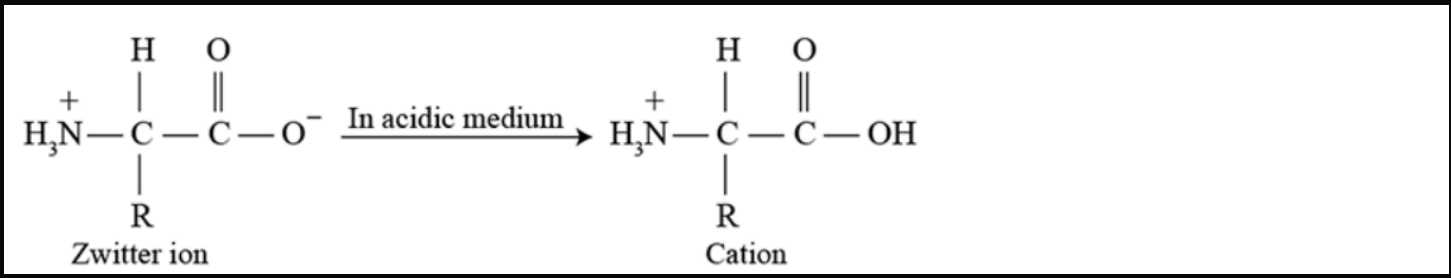
The carbonyl compounds with alpha hydrogen undergoes aldol condensation reaction in the presence of base. The product of aldol condensation is $\alpha,\beta-$unsaturated carbonyl compounds.
The product given is $CH_3\,CH=CH-CHO(\text{but}-2-\text{enal})$
From the structure it is observed that two moles of ethanal are involved in the reaction.

Q.44 Answer:

It exists as cation

Solution:

At low pH (acidic solution) COO^- ion acts as the base and accepts proton to form the cation.



Q.45 Answer:

$\frac{a+b}{2}$

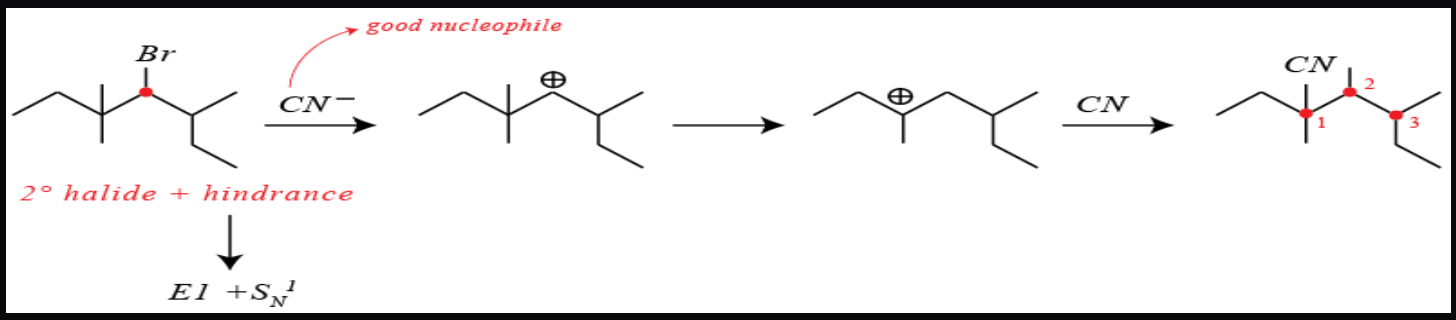
Solution:

$EN=\frac{IE+EA}{2}$ **and** $EA=-E_{ga}$

Q.46 Answer:

—

Solution:



Q.47 Answer:

—

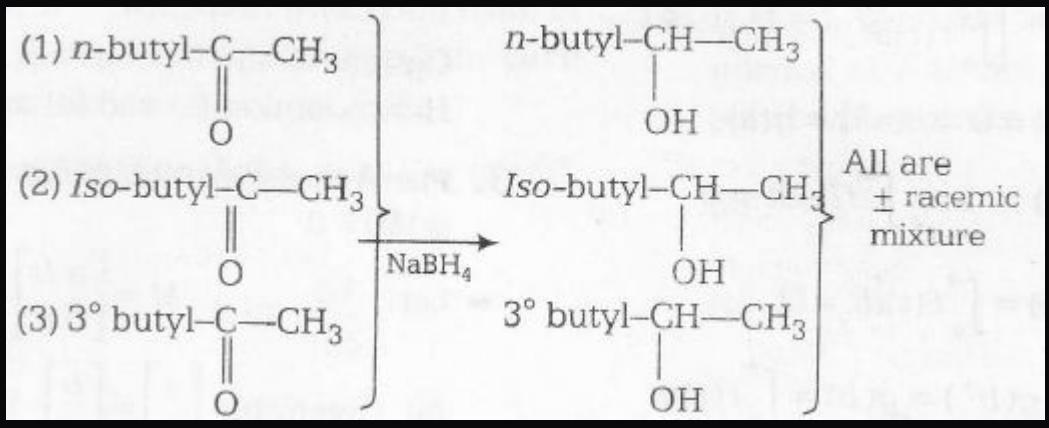
Solution:

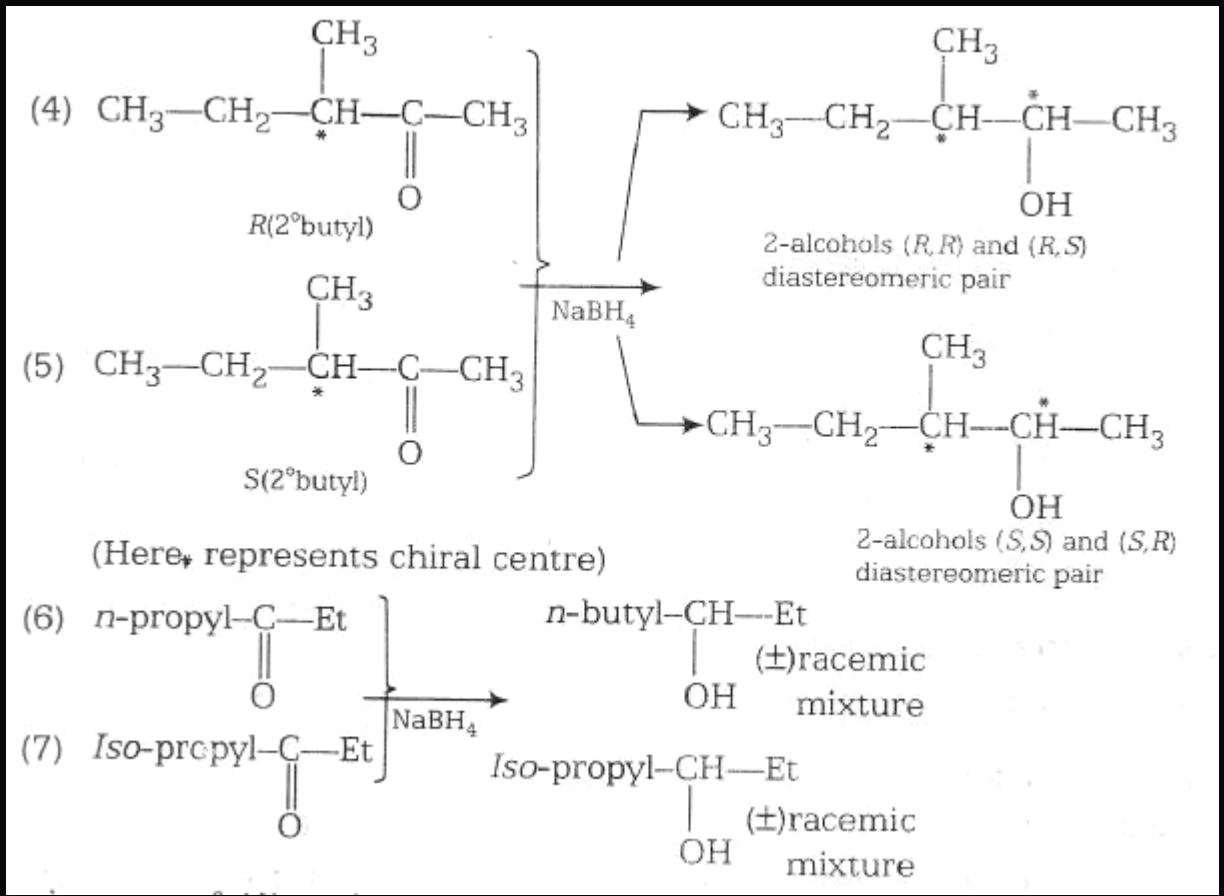
Molecular weight of the ketone is 100

So, molecular formula $= C_6H_{12}O$

Degree of unsaturation $= (6+1) - \frac{12}{2} = 1$

According to question, compound contains ketone group. Since, the compound which contain chiral centre lead to the formation of diastereomer while other produces enantiomers. Various isomers and their possible reduced product are as shown below.



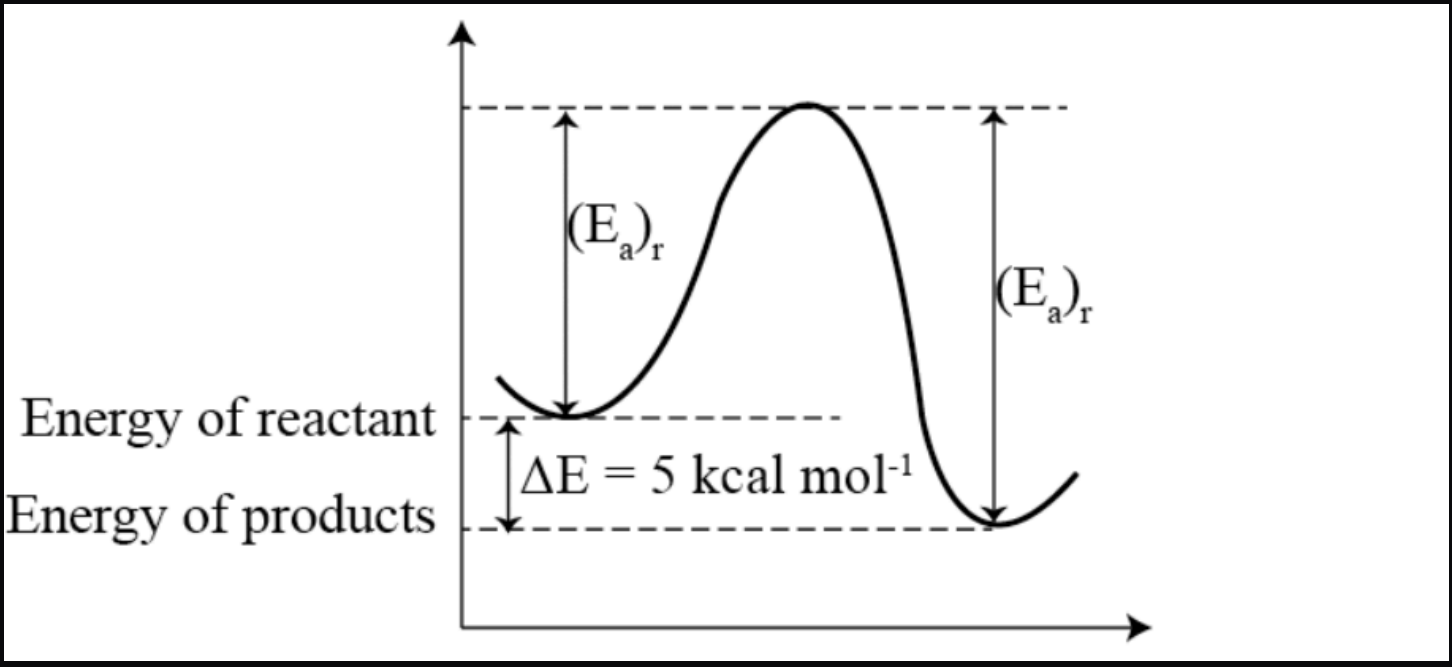


The number of ketones that gives racemic mixture with $NaBH_4$ is 5 as the ketone with chiral center will give diastereoisomer with $NaBH_4$.

Q.48 Answer:

—

Solution:



$(E_a)_r = (E_a)_f + \Delta E = 15 kcal mol^{-1} + 5 kcal mol^{-1}$
$= 20 kcal mol^{-1}$

Q.49 Answer:

—

Solution:

$Cl_2 \rightarrow$ **Greenish yellow** $NO_2 \rightarrow$ **Reddish brown** $O_3 \rightarrow$ **Pale blue**

Q.50 Answer:

—

Solution:

$KE_{final} = E_{photon} + E_{potential} - \phi$

where $\phi =$ **threshold energy for metal surface** $E_{photon} = \frac{hc}{\lambda} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{3000 \times 10^{-10}}$

$\simeq 6.6 \times 10^{-19} J = 4.12 eV$

$E_{potential} = 2 eV$

$\mathrm{KE}_{\text{final}} = \frac{8\times10^{-19}\ \mathrm{J}}{1.6\times10^{-19}\ \mathrm{J}} = 5\mathrm{eV}$

Hence, $\phi = (4.12 + 2 - 5)\mathrm{eV} = 1.125\mathrm{eV}$

Numerical value $= 8 \times 1.125 = 9$

Q.51 Answer:

144

Solution:

Given, $\sigma = 9$

Let a student obtains x **out of** 75**.**

Then his marks out of 100 **are** $\frac{4x}{3}$**.**

$\Rightarrow$ **Each observation is multiplied by** $\frac{4}{3}$

$\Rightarrow$ **New** $\sigma = \frac{4}{3} \times 9 = 12$**,**

$\Rightarrow$ **New Variance** $= \sigma^2 = 144$

Q.52 Answer:

$\frac{1}{2}$

Solution:

$f(x) = -\left(x - \frac{1}{2}\right)^2 + \left(p + \frac{1}{4}\right)$

If $g(x)$ **is discontinuous at** $x = \frac{1}{2}$**, then** $f\left(\frac{1}{2}\right)$ **should be an integer.**

Hence, p **cannot be** $\frac{1}{2}$

Q.53 Answer:

1140

Solution:

The number of solutions (positive integer) of $x + y + z = r$ **is** ${}^{r-1}C_2$

$\therefore$ **The desired number of solutions is**

$\sum_{r=3}^{20} {}^{r-1}C_2$

${}^2C_2 + {}^3C_2 + {}^4C_2 + ... + {}^{19}C_2$

$= \left({}^3C_3 + {}^3C_2\right) + {}^4C_2 + ... + {}^{19}C_2$

$= {}^4C_3 + {}^4C_2 + ... + {}^{18}C_2 + {}^{19}C_2$

$= {}^{20}C_3$ **(Adding in the same way)**

$= \frac{20\times19\times\cancel{18}\,3}{\cancel{3}\times\cancel{2}} = 1140$

Q.54 Answer:

14

Solution:

$2\vec{a} = \vec{b} \times \vec{c} + 2\vec{b}$

Taking dot with $\vec{b}$

$2\left(\vec{a} \cdot \vec{b}\right) = 2\left|\vec{b}\right|^2 \Rightarrow \vec{a} \cdot \vec{b} = 1$

$\Rightarrow \vec{a} = \vec{b}$ **also** $\vec{b} \times \vec{c} = 0$ **(from given equation)**

$\Rightarrow \vec{b} = \vec{c}$ **or** $\vec{b} = -\vec{c}$

So, if $\vec{b} = \vec{c}$**, then** $\left|3\vec{a} + 3\vec{b} + 5\vec{c}\right| = 12$

or if $\vec{b} = -\vec{c}$**, then** $\left|3\vec{a} + 4\vec{b} + 5\vec{c}\right| = 2$

Q.55 Answer:

Always equal to 1

Solution:

$$\int_0^x (f(t))^3 dt = \frac{1}{x^2}\left(\int_0^x f(t)dt\right)^3$$

$$\therefore \int_0^x (f(t))^3 dt = \frac{1}{x^2}(g(x))^3$$

$\therefore \int_0^x (f(t))^3 dt = \frac{1}{x^2}(g(x))^3$

Differentiating $w.r.t.x$,

$\therefore (f(x))^3 = \frac{1}{x^2}3(g(x))^2 g'(x) - \frac{2}{x^3}(g(x))^3$

$\Rightarrow \left(\frac{xg'(x)}{g(x)}\right)^3 - 3\left(\frac{x.g'(x)}{g(x)}\right) + 2 = 0 \left(as \int_0^x f(t)\,dt = g(x) = g'(x)\right)$

$\Rightarrow \frac{xg'(x)}{g(x)} = 1 or -2$

If $\frac{xg'(x)}{g(x)} = 1$

$\Rightarrow xf(x) = \int\limits_0^x f(t)dt$

$\Rightarrow xf'(x) + f(x) = f(x)$

$\Rightarrow f'(x) = 0$

$\Rightarrow f(x) =$**constant** $= f(1) = 1$

$\Rightarrow f(x) = 1$

Or $xf'(x) + f(x) = -2f(x)$

$\frac{f'(x)}{f(x)} = \frac{-3}{x}$

$\log f(x) = -3\log x + \log c$

$\Rightarrow f(x) = c/x^3$

$\Rightarrow f(1) = 1 \Rightarrow f(x) = 1/x^3$ **(decreasing function)**

Q.56 Answer:

$\frac{(\tan x - \cot x)^{2020}}{2020} + c$

Solution:

$I=\int\left(\frac{\sin^2 x-\cos^2 x}{\sin x \cos x}\right)^{2019} \frac{1}{\sin^2 x \cos^2 x} dx$

$=\int(\tan x-\cot x)^{2019}\left(\frac{\sin^2 x+\cos^2 x}{\sin^2 x \cos^2 x}\right) dx$

$=\int(\tan x-\cot x)^{2019}\left(\sec^2 x+\operatorname{cosec}^2 x\right) dx$

Let $\tan x-\cot x=t$

$\Rightarrow\left(\sec^2 x+\operatorname{cosec}^2 x\right) dx=dt$

$\therefore I=\int t^{2019} dt$

$=\frac{t^{2020}}{2020}+c$

$=\frac{(\tan x-\cot x)^{2020}}{2020}+c$

Q.57 Answer:

(2,3]

Solution:

Let, $\vec{a}=2\hat{i}+3\hat{j}$

$\vec{b}=-4\hat{i}+7\hat{j}+5\hat{k} \Rightarrow \vec{b}-\vec{a}=-6\hat{i}+4\hat{j}+5\hat{k}$

$\vec{c}=2\hat{i}+2\hat{j}+\hat{k}$

$\vec{d}=-\hat{i}+8\hat{j}+4\hat{k}$

$\Rightarrow \vec{c}\times\vec{d}=-9\hat{j}+18\hat{k}$

Hence, shortest distance $=\left|\frac{\left(\vec{b}-\vec{a}\right)\cdot\left(\vec{c}\times\vec{d}\right)}{\left|\vec{c}\times\vec{d}\right|}\right|$

$=\left|\frac{-36+90}{9\sqrt{5}}\right|=\frac{6}{\sqrt{5}} \in (2,3]$

Q.58 Answer:

$\left(-\frac{1}{2}, 0\right) \cup (1, \infty)$

Solution:

Since $f'(x)>0$

$\Rightarrow f'(x)$ is always increasing

$g'(x)=2f'\left(2x^3-3x^2\right)\times\left(6x^2-6x\right)+f'\left(6x^2-4x^3-3\right)\left(12x-12x^2\right)$

$=12\left(x^2-x\right)\left(f'\left(2x^3-3x^2\right)-f'\left(6x^2-4x^3-3\right)\right)$

$=12x(x-1)]\left[f'\left(2x^3-3x^2\right)-f'''\left(6x^2-4x^3-3\right)\right)]$

For increasing $g'(x)>0$

Case-I $x<0$ or $x>1$

$\Rightarrow \mathrm{f}\left(2\mathrm{x}^3-3\mathrm{x}^2\right)>\mathrm{f}'\left(6\mathrm{x}^2-4\mathrm{x}^3-3\right)$

$\Rightarrow 2x^3-3x^2>6x^2-4x^3-3$

$(\because f'(x)$ is increasing $)$

$\Rightarrow (x-1)^2\left(x+\frac{1}{2}\right)>0 \Rightarrow x>-\frac{1}{2}$

$\therefore x \in \left(-\frac{1}{2}, 0\right) \cup (1, \infty)$

Case II : If $0<x<1$

$\mathrm{f}'\left(2\mathrm{x}^3-3\mathrm{x}^2\right)<\mathrm{f}'\left(6\mathrm{x}^2-4\mathrm{x}^3-3\right)$

$(x-1)^2\left(x+\frac{1}{2}\right)<0$

$\Rightarrow x<-\frac{1}{2}$, so there is no solution $x<-\frac{1}{2}$

$\Rightarrow$ Hence the values are $x \in \left(-\frac{1}{2}, 0\right) \cup (1, \infty)$

Q.59 Answer:

the minimum value of $f(x)$ is 4

Solution:

We have, $f(x)=\frac{2\left(x^2+1\right)}{[x]}$

When $x \in [1,2)$, then $f(x)=2\left(x^2+1\right) \Rightarrow R_f=[4,10)$

When $x \in [2,3)$, then $f(x)=x^2+1 \Rightarrow R_f=[5,10)$

When $x \in [3,4)$, then $f(x)=\frac{2\left(x^2+1\right)}{3} \Rightarrow R_f=\left[\frac{20}{3}, \frac{34}{3}\right)$

$\therefore R_f=\left[4, \frac{34}{3}\right)$

Clearly, for $x \in [1,2)$ and $x \in [2,3)$ range of $f(x)$ has common elements

$\Rightarrow f(x)$ is many-one

$\Rightarrow f(x)$ is not bijective

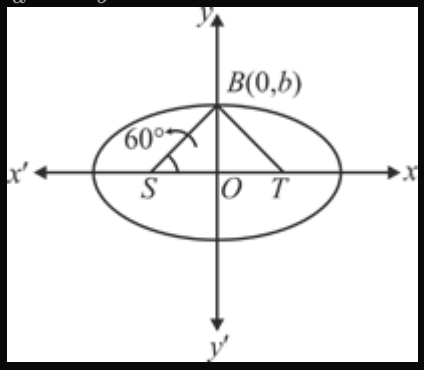
Q.60 Answer:

$\frac{1}{2}$

Solution:

Let the equation of ellipse be

$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$



In ΔSOB,

$$\tan 60^{\circ}=\frac{OB}{OS}$$

$$\Rightarrow \sqrt{3}=\frac{b}{ae}$$

$$\Rightarrow \frac{b}{a}=e\sqrt{3}$$

$\Rightarrow \sqrt{3}=\frac{b}{ae}$

$\Rightarrow \frac{b}{a}=e\sqrt{3}$

Now, $e^2=1-\frac{b^2}{a^2} \Rightarrow e^2=1-3e^2$

$\Rightarrow 4e^2=1 \Rightarrow e=\frac{1}{2}$

Q.61 Answer:

$(-\tan 1, \tan 2)$

Solution:

$\left(\tan^{-1} x - 2\right)\left(\cot^{-1} x - 1 - \frac{\pi}{2}\right) > 0$

$\Rightarrow \left(\tan^{-1} x + 1\right)\left(\tan^{-1} x - 2\right) < 0$

$\Rightarrow -1 < \tan^{-1} x < 2$

$\Rightarrow -\tan 1 < x < \tan 2$

Q.62 Answer:

$\frac{10}{31}$

Solution:

$P(A) = 2(B) \Rightarrow P(B) = \frac{1}{3}P(A) = \frac{2}{3}$

Required Probability $= \frac{\frac{1}{3}\cdot\frac{4}{7}}{\frac{2}{3}\cdot\frac{3}{5}+\frac{1}{3}\cdot\frac{4}{7}} = \frac{10}{31}$.

Q.63 Answer:

$8\sqrt{2}$ units

Solution:

Given, $\mathrm{zw} = |\mathrm{z}|^2 \Rightarrow \mathrm{zw} = \mathrm{z}\bar{\mathrm{z}}$

$\Rightarrow \mathrm{w} = \bar{\mathrm{z}}\ \{\mathrm{z} \neq 0\}$

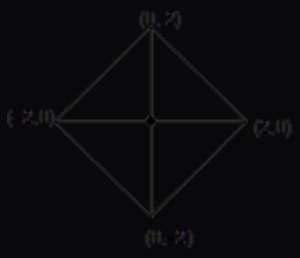
Now, $\left|\mathrm{z} - \bar{\mathrm{z}}\right| + \left|\mathrm{w} + \bar{\mathrm{w}}\right| = 4$

$\Rightarrow \left|\mathrm{z} - \bar{\mathrm{z}}\right| + \left|\mathrm{z} + \bar{\mathrm{z}}\right| = 4$

Let, $\mathrm{z} = \mathrm{x} + i\mathrm{y}$**, then we get,**

$|\mathrm{x}| + |\mathrm{y}| = 2$

which represents a square of side length equal to $2\sqrt{2}$



$\Rightarrow$ **The perimeter of the locus is** $8\sqrt{2}$ **units**

Q.64 Answer:

reflexive, symmetric and not transitive

Solution:

Given, $R = \{(x, y) \in W \times W$ **: the word** x **and** y **have atleast one letter in common}**

Let, $W = \{\text{cat},\ \text{toy},\ \text{you}, \ldots\}$

Clearly, R **is reflexive and symmetric but not transitive.**

[Since, $\text{cat}\, R\, \text{toy},\ \ \text{toy}\, R \text{you} \nRightarrow\ \ \text{cat}\, R\, \text{you}$**]**

Q.65 Answer:

30

Solution:

Let $P = \frac{\left(a^8+4a^4+1\right)\left(b^4+3b^2+1\right)\left(c^2+2c+2\right)}{a^4b^2}$

$$= \left(a^4 + 4 + \frac{1}{a^4}\right)\left(b^2 + 3 + \frac{1}{b^2}\right)\left\{(c+1)^2 + 1)\right\}$$

$$\because = a^4 + 4 + \frac{1}{a^4} \geq 6, b^2 + 3 + \frac{1}{b^2} \geq 5 \text{ and } (c+1)^2 + 1 \geq 1$$

$$\therefore \qquad P \geq 6 \cdot 5 \cdot 1 = 30 \Rightarrow P \geq 30$$

Hence, the required minimum value is 30 .

Q.66 Answer:

$b \times a = 7$

Solution:

$x + y - 2 + \lambda(2x - y + 1) = 0$

$\Rightarrow 1 + 1 = (1 + 2\lambda)^2 + (1 - \lambda)^2$

$1 + 1 = 1 + 4\lambda + 4\lambda^2 + 1 - 2\lambda + \lambda^2$

$\Rightarrow 5\lambda^2 + 2\lambda = 0$

$\Rightarrow \lambda \neq 0,\ \lambda = \frac{-2}{5}$

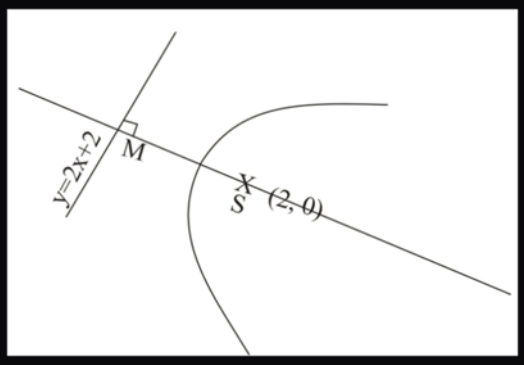
$5(x + y - 2) - 2(2\,x - y + 1) = 0$

$\Rightarrow x + 7y - 12 = 0.$

Q.67 Answer:

$\frac{\sqrt{5}}{12}$

Solution:



Length of $\text{LR} = 2(\text{SM})$

$= 2\ \frac{|2(2)+2-0|}{\sqrt{2^2+1}} = \frac{12}{\sqrt{5}}$

$\Rightarrow \frac{1}{\mathrm{K}} = \frac{12}{\sqrt{5}} \Rightarrow \mathrm{K} = \frac{\sqrt{5}}{12}$

Q.68 Answer:

$\frac{1}{5}$

Solution:

$$\mathrm{a_{ij}} = \begin{cases} 3, & \mathrm{i = j} \\ 0, & \text{otherwise} \end{cases}$$

$$A = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$|A| = 3(9-0)+0+0$

$|\mathrm{A}| = 27$

$|adj(adjA)| = |A|^{(n-1)^2}$

$= (27)^{2^2}$

$= (27)^4$

$\frac{|adj(adjA)|}{5} = \frac{(27)^4}{5}$

$\left\{\frac{|adj(adjA)|}{5}\right\} = \frac{|adj(adjA)|}{5} - \left[\frac{|adj(adjA)|}{5}\right]$

$= \frac{(27)^4}{5} - \left[\frac{(27)^4}{5}\right]$

$= \frac{531441}{5} - 106288$

$= \frac{531441-531440}{5}$

$= \frac{1}{5}$

Q.69 Answer:

$2p-1$

Solution:

Given $A = \begin{bmatrix} a & b \\ c & a \end{bmatrix}, a, b, c \in \{0, 1, 2, \ldots, (p-1)\}$

If A **is skew - symmetric matrix, then** $a = 0, b = -c$

$\therefore |A| = -b^2$

Thus, p **divides** $|A|$ **only when** $b = 0$ **...(1)**

Again, If A **is symmetric matrix, then** $b = c$ **and** $|A| = a^2 - b^2$

Thus, p **divides** $|A|$ **if either** p **divides** $(a-b)$ **or** p **divides** $(a+b)$

p **divides** $(a-b)$**, only when** $a = b$

i.e., $a = b \in \{0, 1, 2, \ldots, (p-1)\}$ **...(2)**

i.e., p **choices**

p **divides** $(a+b)$**then** $a+b=p$**, hence** (a,b)**can take values** $(0,0), (1,1) \ldots (p-1, p-1)$

$\Rightarrow$ p **choices, including** $a=b=0$ **included in equation (1)**

$\therefore$ **Total number of choices are** $(p+p-1) = 2p-1$

Q.70 Answer:

$\frac{1}{6}(1+3\sqrt{5})$

Solution:

$$\int_{-1}^{1} \{x^2+x-3\}\,dx = \int_{-1}^{1} \{x^2+x\}\,dx$$

$$= \int_{-1}^{1} \left(x^2+x-\left[x^2+x\right]\right)dx$$

$$= \left[\frac{x^3}{3}+\frac{x^2}{2}\right]_{-1}^{1} - \int_{-1}^{0}\left[x^2+x\right]dx$$

$$\int_{0}^{\frac{\sqrt{5}-1}{2}}\left[x^2+x\right]dx$$

$$- \int_{(\sqrt{5}-1)_2}^{1}\left[x^2+x\right]dx$$

$$= \frac{2}{3}+1-0-1\left(1-\frac{\sqrt{5}-1}{2}\right)$$

$$= \frac{2}{3}+\frac{\sqrt{5}-1}{2}$$

$$= \frac{1}{6}(1+3\sqrt{5})$$

Q.71 Answer:

—

Solution:

Given,

$$a_0 = 0, a_n = 3a_{n-1} + 1 (n \geq 1)$$

$$a_{2010} = 3a_{2009} + 1$$

$$= 3\left(3a_{2008} + 1\right) + 1 = 3^2 \cdot a_{2008} + 3 + 1$$

$$a_{2010} = 3^2\left(3 \cdot a_{2007} + 1\right) + 3 + 1 = 3^3 \cdot a_{2007} + 3^2 + 3 + 1$$

$$= 3^3\left(3 \cdot a_{2006} + 1\right) + 3^2 + 3 + 1$$

$$= 3^4 \cdot a_{2006} + 3^3 + 3^2 + 3^1 + 1$$

$$a_{2010} = 3a_{2009} + 1$$

$$= 3\left(3a_{2008} + 1\right) + 1 = 3^2 \cdot a_{2008} + 3 + 1$$

$$a_{2010} = 3^2\left(3 \cdot a_{2007} + 1\right) + 3 + 1 = 3^3 \cdot a_{2007} + 3^2 + 3 + 1$$

$$= 3^3\left(3 \cdot a_{2006} + 1\right) + 3^2 + 3 + 1$$

$$= 3^4 \cdot a_{2006} + 3^3 + 3^2 + 3^1 + 1$$

Similarly,

$$a_{2010} = 3^{2010}a_0 + \left(3^{2009} + 3^{2008} + \ldots + 3 + 1\right)$$

$$(a_0 = 0)$$

$$(a_0 = 0)$$

So,

$$a_{2010} = 1 + 3 + 3^2 + \ldots + 3^{2009}$$

$$= \left(\frac{3^{2010} - 1}{2}\right)$$

$$= \frac{\left(3^5\right)^{402} - 1}{2} = \frac{(243)^{402} - 1}{2}$$

$$= \frac{(1 + 242)^{402} - 1}{2}$$

$$= \frac{\left({}^{402}C_0 + {}^{402}C_1 242 + {}^{402}C_2 (242)^2 + \ldots\right) - 1}{2}$$

$$= \frac{{}^{402}C_1 242 + {}^{402}C_2 (242)^2 + \ldots}{2}$$

$$= \frac{2\left[{}^{402}C_1 121 + {}^{402}C_2 2(121)^2 + \ldots\right]}{2}$$

$$= \left(\frac{3^{2010} - 1}{2}\right)$$

$$= \frac{\left(3^5\right)^{402} - 1}{2} = \frac{(243)^{402} - 1}{2}$$

$$= \frac{(1 + 242)^{402} - 1}{2}$$

$$= \frac{\left({}^{402}C_0 + {}^{402}C_1 242 + {}^{402}C_2 (242)^2 + \ldots\right) - 1}{2}$$

$$= \frac{{}^{402}C_1 242 + {}^{402}C_2 (242)^2 + \ldots}{2}$$

$$= \frac{{}^{402}C_1 242 + {}^{402}C_2 (242)^2 + \ldots}{2}$$

$$= \frac{2\left[{}^{402}C_1 121 + {}^{402}C_2 2(121)^2 + \ldots\right]}{2}$$

$= \left[{}^{402}C_1 121 + {}^{402}C_2 2(121)^2 + \ldots\right]$

$= 121\left[{}^{402}C_1 + {}^{402}C_2 \times 2 \times 121 + \ldots\right]$

$= 11 \times 11\left[{}^{402}C_1 + {}^{402}C_2 \times 2 \times 121 + \ldots\right]$

$= 121\left[{}^{402}C_1 + {}^{402}C_2 \times 2 \times 121 + \ldots\right]$

$= 11 \times 11\left[{}^{402}C_1 + {}^{402}C_2 \times 2 \times 121 + \ldots\right]$

Which is divisible by 11.

Q.72 Answer:

—

Solution:

Given equation: $e^x = kx^2$

$\Rightarrow k = \frac{e^x}{x^2}$

Suppose,

$f(x) = \frac{e^x}{x^2}$

$\Rightarrow \ f\prime(x) = \frac{x^2 e^x - e^x.2x}{x^4} = \frac{e^x x(x-2)}{x^4}$

$\Rightarrow \ f\prime(x) = \frac{(x-2)e^x}{x^3}$

For critical points, $f\prime(x) = 0$

or $\frac{(x-2)e^x}{x^3} = 0$

Now, $x = 2$ **is a minima as** $f\prime(2^+) > 0$ **and** $f\prime(2^-) < 0$.

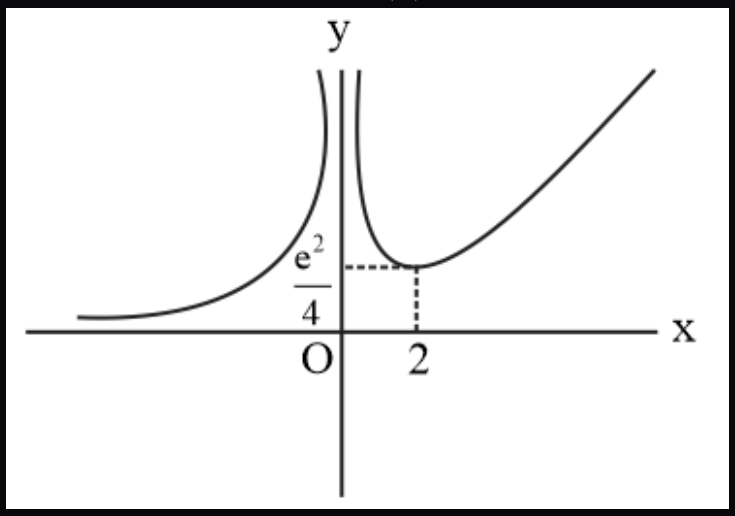
Now, $\lim_{x\to-\infty} f(x) = \lim_{x\to-\infty} \frac{e^x}{x^2} = 0$

$\lim_{x\to0^-} f(x) = \lim_{x\to0^-} \frac{e^x}{x^2} \to \infty$

$\lim_{x\to0^+} f(x) = \lim_{x\to0^+} \frac{e^x}{x^2} \to \infty$

$\lim_{x\to\infty} f(x) = \lim_{x\to\infty} \frac{e^x}{x^2} \to \infty$

The graph of the function $f(x)$**:**



For three distinct solutions,

$k > \frac{e^2}{4}$; $k \in I$ **as in that case it will definitely have three points of intersection with the curve, so,**

$k > 1.84472..$

$\therefore k_{min} = 2$.

Q.73 Answer:

—

Solution:

$x \tan \frac{y}{x} - y \sec^2 \left(\frac{y}{x}\right) + x \sec^2 \left(\frac{y}{x}\right) \frac{dy}{dx} = 0$
$\Rightarrow \frac{dy}{dx} = \frac{y}{x} - \frac{1}{2} \sin \left(\frac{2y}{x}\right)$
Put $y = vx$
$\frac{dy}{dx} = v + x \frac{dv}{dx}$
$v + x \frac{dv}{dx} = v - \frac{1}{2} \sin 2v$
$2 \operatorname{cosec} 2v dv = -\frac{dx}{x}$
$\ln(\tan v) = -\ln x + \ln C$
$\Rightarrow x \tan \frac{y}{x} = C$
$x = 2, y = \frac{\pi}{4}$
$x \tan \frac{y}{x} = C$
$2 \tan \frac{\pi}{8} = C$
$C = 2(\sqrt{2} - 1) = 2(1 \cdot 414 - 1) = 0.828$
$\Rightarrow 0.82$
or 0.83

Q.74 Answer:

—

Solution:

$27^{40} = \left(3^3\right)^{40} = 3^{120}\ 3^{119} = (4-1)^{119} = {}^{119}C_0 4^{119} - {}^{119}C_1 4^{118} + {}^{119}C_2 4^{117} - \ldots\ldots\ldots + {}^{119}C_{118} 4 - {}^{119}C_{119} = 4k - 1 \Rightarrow 3^{120} = 3(4\mathrm{k} - 1) = 12k - 3 = 12(k-1) + 9$ **So, the required remainder is** 9**.**

Q.75 Answer:

—

Solution:

adding all wehave
$6(a+b+c+d+e) = 186$
$a+b+c+d+e = 31 \ldots . (1)$
from
$(1)\ \&\ (2) \Rightarrow a - b = -b$
$(1)\ \&\ (3) \Rightarrow a - c = -18$
$(1)\ \&\ (4) \Rightarrow a - d = -42$
$(1)\ \&\ (\mathrm{s}) \Rightarrow a - c = -90$
$(2)\ \&\ (3) \Rightarrow b - c = -12$
$(2)\ \&(4) \Rightarrow \quad b - d = -36$
$(2)\ \&\ (5) \Rightarrow b - e = -84$
$(3)\ \&\ (9) \Rightarrow c - d = -24$
$(3)\ \&\ (5) \Rightarrow \quad c - e = -72$
$(4)\ \&(5) \Rightarrow d - e = -48$